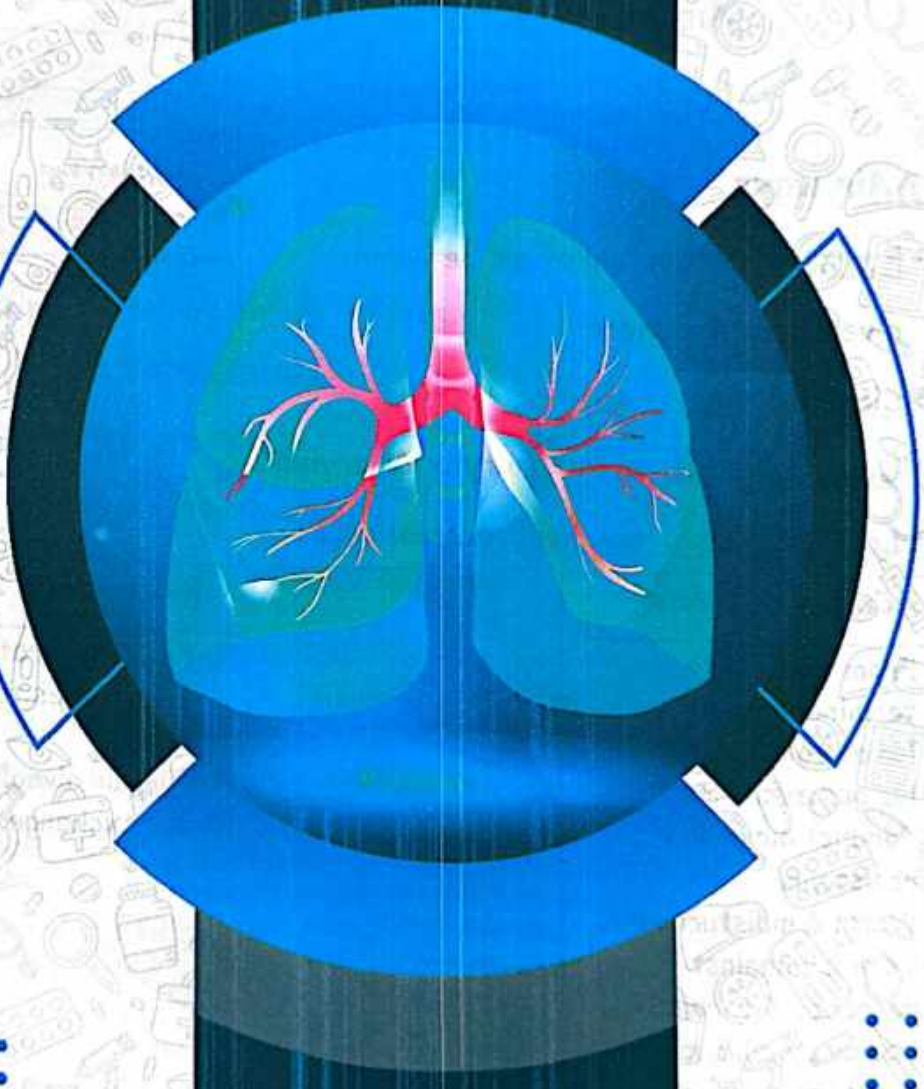
2026

PHSYOLOGY

RESPIRATORY



FOR 2ND YEAR

MEDICAL STUDENTS

DR. ABDELHAMID

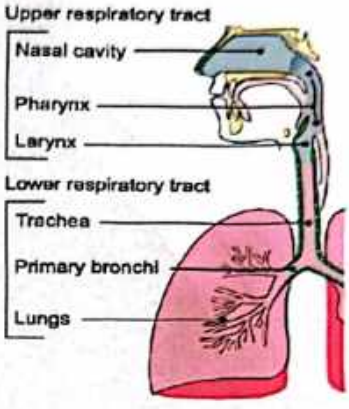
Respiratory Physiology 2026

Respiratory physiology

Structure of respiratory system

1- Air ways

2- Lungs



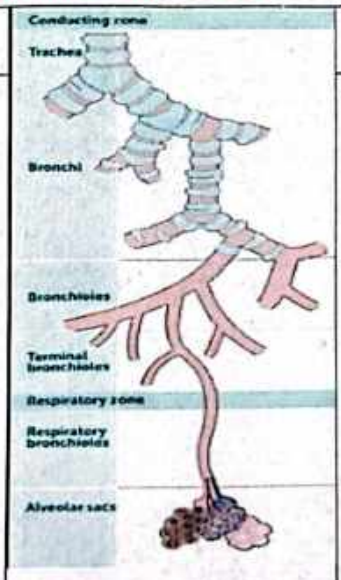
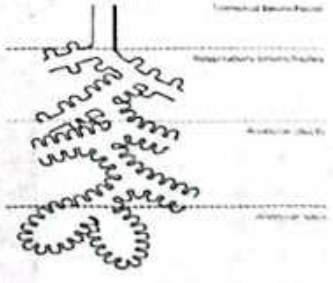
1) Air ways

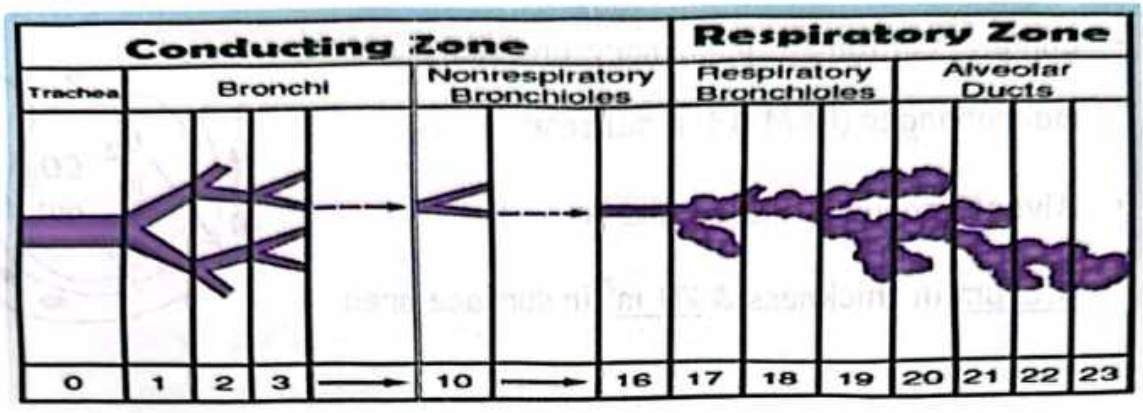
☺ **Function:** conduction of air between alveoli & atmosphere

☺ **Classification:**

a) **Upper airways:** nose, pharynx & larynx

b) **Lower airways:** tracheobronchial tree:

Conducting zone		Respiratory zone
From trachea to terminal bronchioles No alveoli No gas exchange Conduct air at low resistance Warm & moistures air Defense against microbes & foreign bodies by cilia, mucous & phagocytosis Helps in phonation		From respiratory bronchioles to alveoli Contain alveoli gas exchange 



2) Lungs

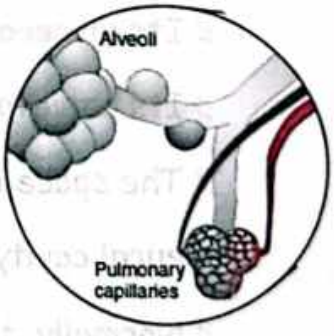
Right lung (3 lobes) & left lung (2 lobes)

Structure of lungs:



- Branched airways
- Pulmonary blood vessels
- Elastic connective tissue
- Alveoli: Thin wall, inflatable, grape like sacs

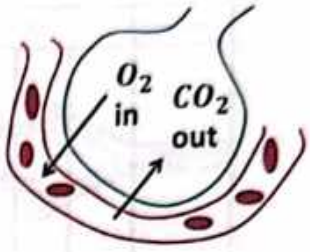
Characters of alveoli:



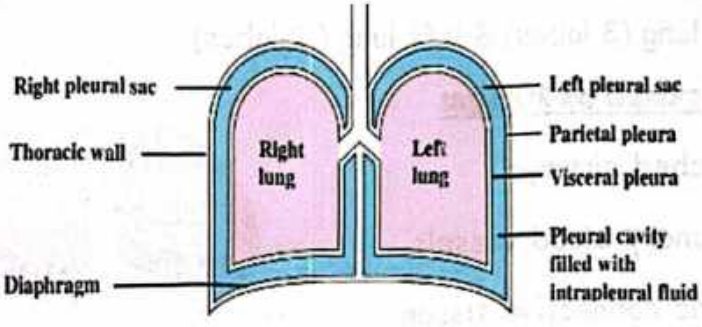
- 300 $x10^6$ in number
- Each alveolus is surrounded by ~ 1000 capillaries.
- Lined by type I epithelial cells (single layer of flat cells)
- Contain type II cells (granular pneumocytes), that secrete surfactant.

- Lumen contains pulmonary alveolar macrophages (PAMs) for defense
- Alveolo-capillary membrane is:

  0.5 µm in thickness & 70 m^2 in surface area



Pleural sac



A double walled closed sac.

Its visceral layer covers lung.

Its parietal layer lines the chest (thoracic wall).

The space in between the 2 layer is the pleural space (or pleural cavity)

Normally, the pleural cavity contains no air & it contains only few milliliters of intrapleural fluid, which is oily fluid (Lymph-Like) for lubrication & to prevent friction.

Function of respiratory system

1) Respiratory function:

Gas exchange

- The 1ry function ← main function.
- Gas exchange between alveolar air & pulmonary capillary blood
- For: O_2 supply to tissues & removal of CO_2 from tissues.

2) Non-respiratory functions:

a) General functions:

1) Regulation of body temperature (heat loss)
2) Regulation of acid-base balance (by regulating CO_2 loss)
3) Venous & lymphatic return (thoracic pump mechanism) ← من تحت لفوق عكس الجاذبية
4) Voice production
5) Smell

b) Metabolic & endocrine functions

1) **Synthesis of:** surfactant (local), histamine & prostaglandins [PG] (to blood)
2) **Removal of:** serotonin, some PG, A.Ch & N.Epi from blood
3) **Conversion of** angiotensin I to angiotensin II by angiotensin converting enzyme [ACE]

4) **Inactivation of** bradykinin & other chemicals

5) **Lysis of** blood clot (fibrinolytic activity) by secretion of tissue plasminogen activator

6) **Filtering of** blood from blood clots, chemicals & tissue debris before returning back to the heart

7) **Vascular reservoir**, by accommodation of large blood volume (from 50-70 → 500-1000 mL) during exercise & deep inspiration

Mechanism: distension of capillaries & recruitment (↑opening of capillaries

c) Defensive functions:

1) **Ig A** is present in bronchial secretion

2) **Large hair** (vibrissae) in nostrils trap (prevent passage) of large foreign particles (> 10 μ)

3) Smaller foreign particles (2-10 μ) stick to **bronchial mucous** & moved away by **ciliary movement** (ciliary escalator mechanism)

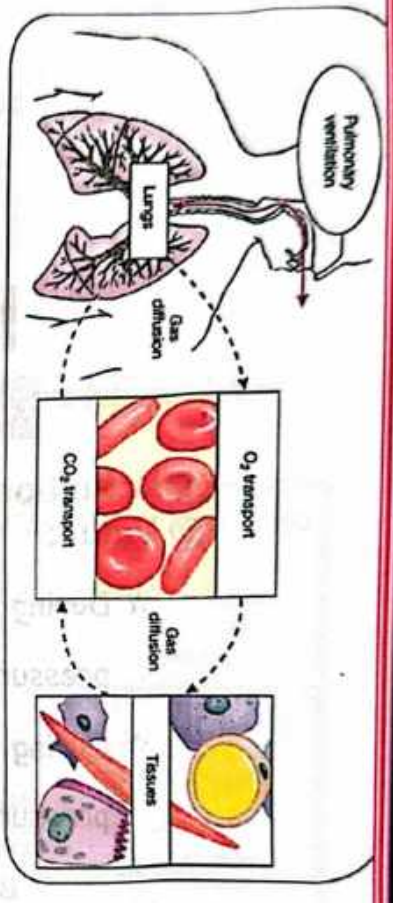
4) Very small particles (less than 2 μ) reach the alveoli & phagocytosed by **pulmonary alveolar macrophages** (PAMs).

Respiration

Respiration is divided **physiologically** into 2 major processes:

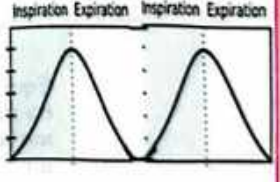
a) External respiration

1) **Pulmonary ventilation** (by the respiratory system): mass flow of air between lungs & atmosphere down pressure gradient.
2) **Exchange of gases** (by the respiratory system): by diffusion between alveolar air & pulmonary capillary blood.
3) **Transport of gases** (by the CVS): between lungs & tissues.
4) **Diffusion of gases** (by the CVS): between blood & tissues through systemic capillaries



b) Internal respiration = cellular respiration

= Intracellular metabolic processes in mitochondria by using O_2 & producing CO_2



☎ Respiration is divided **MECHANICALLY** into cycles.

☎ Each Respiratory cycle = inspiration + expiration + pause

☎ Normally: 12-16 cycles/min.

Respiratory pressures

⌘ Intra-alveolar pressure

⌘ Intra-pleural pressure

⌘ Trans-pulmonary pressure

⌘ Trans-thoracic pressure

1. Intra alveolar pressure

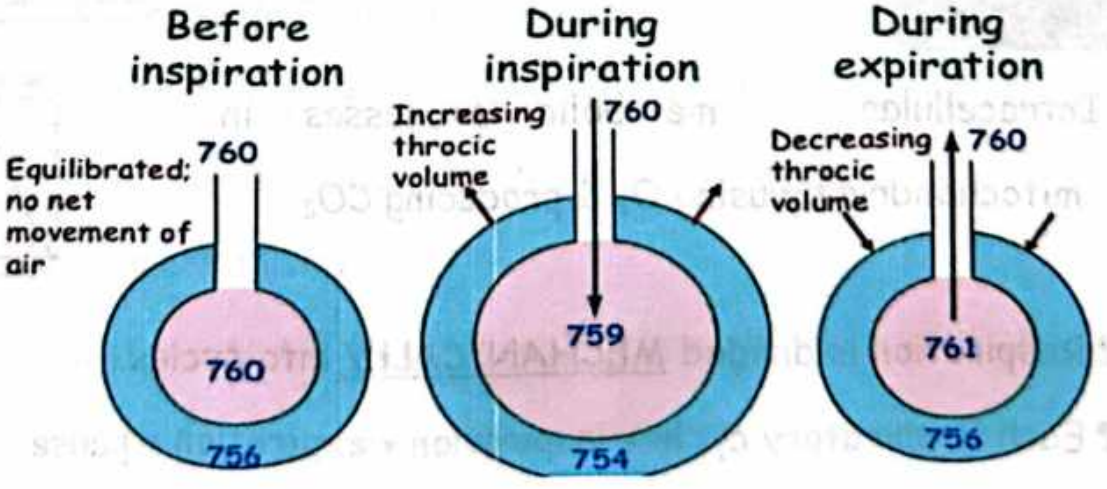
= intrapulmonary pressure

@ **Definition:** it is the pressure inside alveoli

@ **Value:**

☆ **During inspiration**→ it decreases below atmospheric pressure → 759 = -1 mmHg → air rushes in

☆ **During expiration**→ it increases above atmospheric pressure → 761 = + 1 mmHg → air rushes out

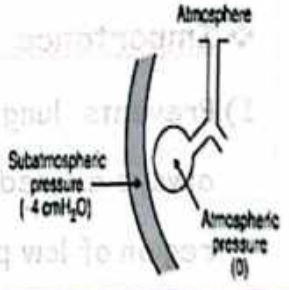


- Intraalveolar pressure = O mmHg (= 760 mmHg) at the end of inspiration & at the end of expiration
- ***Intraalveolar pressure always equalizes itself with atmospheric pressure*** because they are connected

2.Intrapleural pressure = IPP = Intrathoracic pressure

@ **Definition:** it is the pressure between the 2 layers (visceral & parietal) of pleura.

- Normally; the intrapleural space contains few milliliters of lymph- like fluid for lubrication of movements & normally there is no air
- During quiet breathing, the intrapleural pressure is ***ALWAYS -ve (subatmospheric)*** & cannot equilibrate with intraalveolar or atmospheric pressures



@ Normal values:

★ At the end of normal expiration = - 4 mmHg (= 756 mmHg)

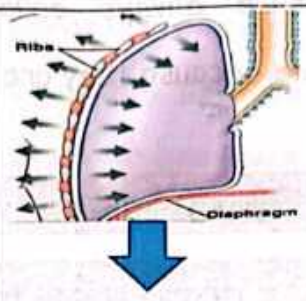
★ At the end of normal inspiration = - 6 mmHg (= 754 mmHg)

★ At deep inspiration -30 mmHg

★ It is always -ve (subatmospheric)

❖ Causes of negativity of IPP:

The continuous tendency of:

Chest wall to expand to its uncompressed size before birth BECAUSE: # It is compressed		Lung to recoil to its unstretched size before birth BECAUSE: # It is stretched # Surface tension of fluid lining alveoli
A vacuum occurs → ↓in pressure		

❖ Measurement of IPP:

By: intraoesphaged balloon connected to a sensitive manometer

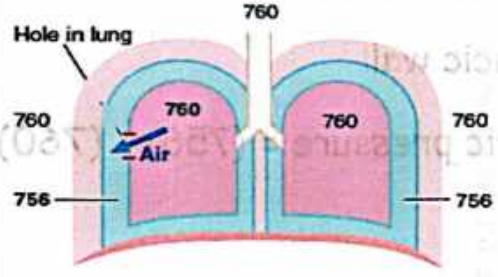
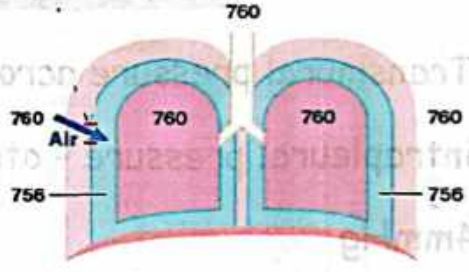
❖ Importance of intapleural pressure:

1) **Prevents lung collapse**: because the alveolar pressure always exceeds IPP. So, the lung tends to expand to the region of low pressure
2) **Helps respiratory movements**: the pressure inside alveoli is higher than the pressure outside them
3) **↓ work of breathing**
4) **Helps venous & lymphatic return** against gravity
5) **Index of lung elasticity**: the more its -vity, the more the lung elasticity

Pneumothorax

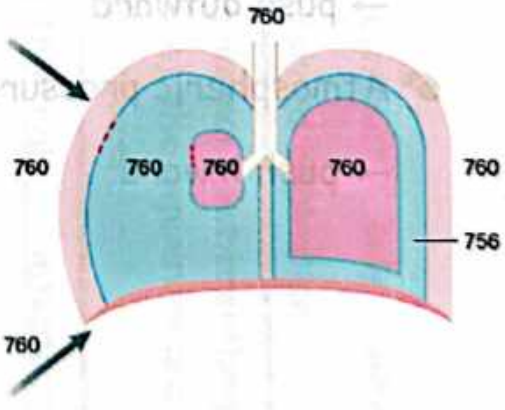
♫ **Definition:** air in the intrapleural space → ↑ intrapleural pressure → IPP = atmospheric Pressure = intraalveolar pressure → loss of transmural pressure Gradient → lung collapse

♫ Types & causes:

Internal (closed = spontaneous)	External (opened= traumatic)
due to rupture of alveoli into the pleural space.  e.g. severe cough in T.B.	due to injury of chest wall  e.g. trauma, stab wound.

♫ Effects:

1. Collapse of lung = atelectasis
2. Distension of chest wall
3. Shift of mediastinum (= midline structures) to the opposite side



3. Transpulmonary pressure

=Transmural pressure across lungs

= alveolar pressure - pleural pressure = (760) - (756) = 4mmHg

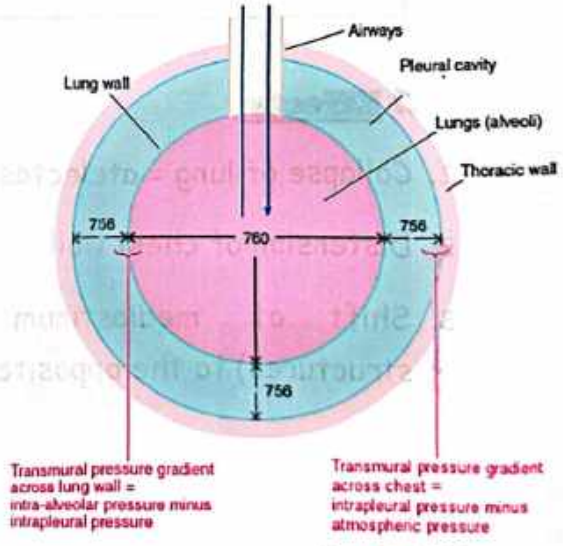
- It is the distending pressure of alveoli (lungs)
- Intraalveolar pressure (760) → push outward
- Intrapleural pressure (756) → push inward

4. Transthoracic pressure

= Transmural pressure across thoracic wall

= intrapleural pressure - atmospheric pressure = (756) - (760) = - 4mmHg

- It is the compressing pressure of the chest
- Intrapleural pressure (756) → push outward
- Atmospheric pressure (760) → push inward



Respiratory muscles

Main inspiratory Muscles	Accessory inspiratory Muscles	Expiratory muscles
♫ work during normal inspiration	♫ work only during forced inspiration	♫ contract only with forced expiration

2) Diaphragm:

- Supplied by phrenic nerve. (C3,4,5)
- Its Contraction → its descent → ↑ vertical diameter of chest wall
- It causes 75% of the increase in chest size

1) External intercostal muscles:-

- *Supplied by intercostal nerves*
- *Their contraction causes:*

- Elevation (up) of ribs → ↑ transverse diameter of chest

- Eversion (forward) of ribs → ↑ anteroposterior diameter of chest

1) Sternomastoid muscles:

Lifts the sternum

2) Scalenus muscles:

Lifts 1st & 2nd ribs

3) Serratus anterior:

Lifts many of ribs

During normal quiet breathing (eupnoea), the expiration is a passive process

1) Internal intercostal muscles:

Their contraction →: depression & inversion of ribs → ↓ chest size

2) Abdominal muscles:

Their contraction →: ↑ intraabdominal pressure → upward movement of diaphragm → ↓ chest size

Scanned
CamSc

Mechanism of quiet breathing

Inspiration	Expiration
◷ Active	◷ Passive
◷ By contraction of main inspiratory muscles	◷ By relaxation of main inspiratory muscles & elastic recoil of lungs
◷ Contraction of diaphragm & external intercostal muscles ↓	◷ Relaxation of diaphragm & external intercostal muscles ↓
◷ enlargement of thoracic cavity ↓	◷ ↓ volume of thoracic cavity ↓
◷ ↓ IPP to 754 mmHg	◷ ↑ IPP to 756 mmHg
◷ ↑ transpulmonary pressure gradient ↓	◷ ↓ transpulmonary pressure gradient ↓
◷ stretch of lung ↓	◷ ↓ lung volume by elastic recoil ↓
◷ ↓ intra alveolar pressure to 759 mmHg ↓	◷ ↑ intra alveolar pressure to 761 mmHg ↓
◷ flow of air to alveoli till intra alveolar pressure become 760 mmHg	◷ flow of air out of alveoli till intra alveolar pressure become 760 mmHg

Mechanism of deep breathing

Deep inspiration

Active

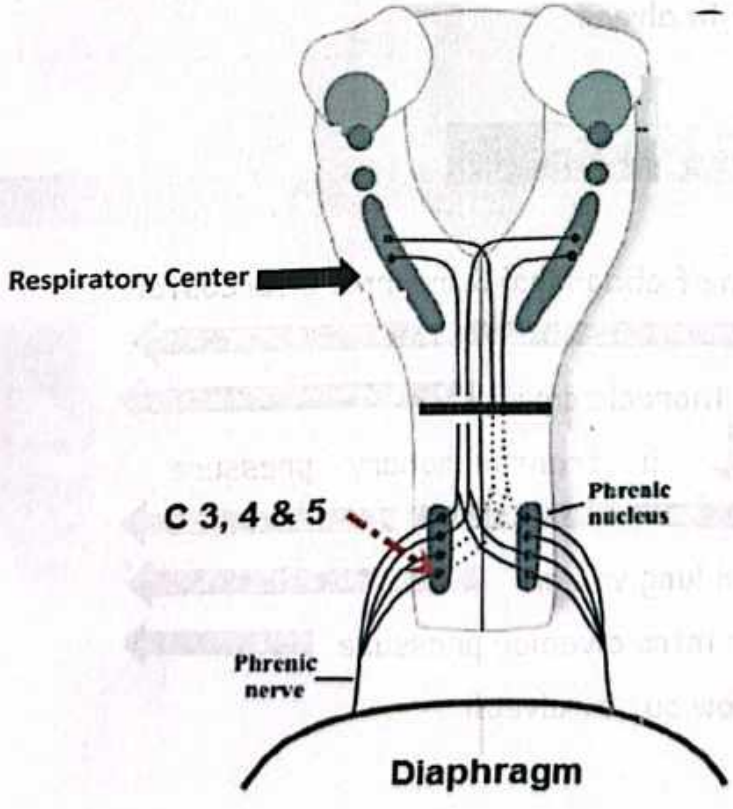
- Strong contraction of diaphragm & external intercostal muscles + contraction of accessory inspiratory muscles →
- more ↑ in thoracic cavity →
- more ↑ in transpulmonary pressure gradient →
- more stretch of lungs →
- more ↓ in intra alveolar pressure →
- more air flow to alveoli

Deep expiration

Active

- Contraction of abdominal & internal intercostal muscles →
- more ↓ in thoracic cavity →
- more ↓ in transpulmonary pressure gradient →
- more ↓ in lung volume →
- more ↑ in intra alveolar pressure →
- more air flow out of alveoli

- Cutting of spinal cord above C3 → paralysis of diaphragm → Death.
- Paralysis of external intercostal muscles alone → no marked effect on normal breathing
- Cutting of spinal cord below C5 → paralysis of external intercostal muscles + paralysis of all body skeletal muscles + *no paralysis of diaphragm* → no marked effect on normal breathing.



- **Euopnea:** normal quiet resting breathing
- **Dyspnea:** difficult breathing
- **Orthopnea:** dyspnea in recumbent (lying down) position
- **Apnea:** temporary stoppage of breathing
- **Hyperpnea:** ↑ rate or death of breathing due to ↑ body metabolic needs
- **Tachypnea:** ↑ rate & ↓ depth (shallow) breathing
- **Hyperventilation:** ↑ ventilation more than metabolic needs → ↓ blood CO_2 → alkalosis
- **Hypoventilation:** ↓ ventilation more than metabolic needs → ↑ blood CO_2 → acidosis
- **Hypocapnia:** ↓ CO_2 in arterial blood
- **Hypercapnia:** ↑ CO_2 in arterial blood
- **Respiratory arrest:** permanent cessation of breathing
- **Suffocation:** inability to breath oxygenated air → O_2 deprivation
- **Asphyxia:** O_2 starvation in tissues due to:
  - ↓O_2 in air
  - Respiratory disease
  - Tissue disease

Airway resistance

⌘ Flow of air = $\frac{\Delta P}{R}$

⌘ Flow of air ∝ Δ P (pressure gradient between atmospheric and intraalveolar pressures)

⌘ Flow of air ∝ $\frac{1}{R}$ (resistance of airways)

- Normally, **resistance is very small.**

  So, a 1 mmHg pressure difference causes movement of 500mL of air.

- **So, The Δ P (pressure gradient) is the primary factor that determine flow of air.**

Resistance

✹ $R = \frac{8 \eta l}{\pi r^4}$

η = viscosity L = length r = radius

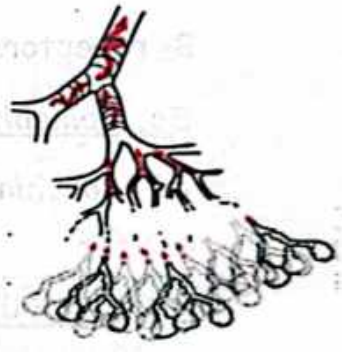
So, ♠ R ∝ viscosity & length ♠ $R \propto \frac{1}{r^4}$

💣 Viscosity of air & length of airways are constant.

💣 **The r^4 (radius of airways) is the primary factor that determines airway resistance.**

Distribution of airway resistance

- **Resistance is:**
  - ✈ **High** in a single terminal bronchiole than in a single labor bronchus
  - ✈ **Highest** in medium size bronchi (small Total Cross Sectional Area [TCSA])
  - ✈ **High** in pipe smoking (↑ length)
  - ✈ **Low** in mouth breathing (↓ length)
  - ✈ **Low** in tracheostomy (↓ length)
  - ✈ **Lowest** in small airways (terminal & respiratory bronchioles) due to:
    1) Parallel arrangement of small airways
    2) Very large TCSA



This is important to: give sufficient time for gas exchange

- **Air flow rate** = amount of air flow through a particular device / unit time
- **Air velocity** = distance travelled by air / unit time

Factors affecting resistance

= factors affecting radius of air ways

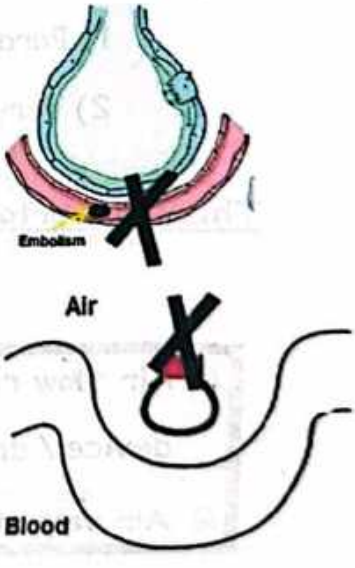
1) Nervous factors:

Affect tone of bronchial smooth muscles

- **Parasympathetic** (the most important, in relaxed state) → A.Ch. → Bronchoconstriction → ↑ resistance.
- **Sympathetic** (little effect, in stress states) → Nor.Epi → B_2 receptors → bronchodilatation → ↓ resistance.

  So, epinephrine (sympathetic effect) is used in treatment of bronchial asthma.

2) Local control (By alveolar CO_2)



- ↓ **CO_2 in pulmonary capillaries** (due to embolism or thrombosis → ↓ blood flow than air flow → direct bronchoconstriction → ↑ airway resistance & ↓ air flow
- ↑ **CO_2 in alveolar air** (due to more blood flow than air flow) → relaxation of bronchial smooth muscles → bronchodilation → ↓ air way resistance & ↑ air flow

3) Pathological actors: -

(Chronic Obstructive Pulmonary Diseases [COPD])

All COPD are:

Characterized by	Triggered by
♣ ↑ air way resistance ♣ Need high pressure (work of muscles) to breathe ♣ Difficult expiation	♠ Smoking ♠ Polluted air ♠ Allergens (dust & pollen)

1) Bronchial asthma:



= disease in which smooth muscles of airways contract in attacks → ↑ airway resistance

Causes:

⊙ Allergy

⊙ Excess mucous secretion & oedema and thickening of airways

⊙ Hyper response of smooth muscles to allergens

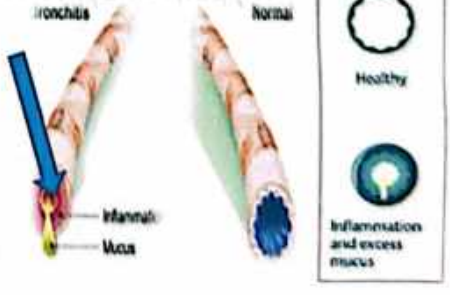
2) Chronic bronchitis:

Inflammation of lower airway →

⊙ Excess mucous secretion

⊙ Thickening of airways

⊙ Secretion of histamine



→ ↑ resistance

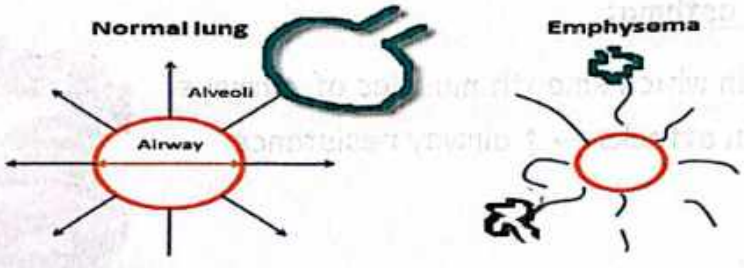
3) Emphysema:

= irreversible collapse of alveoli = damage of alveolar walls & ↓ elastic fibers. → ↓ lateral traction → ↑ breakdown of alveolar walls & ↑ collapse of small airways →

Ω ↑ Airway resistance

Ω ↓ Surface area of gas exchange

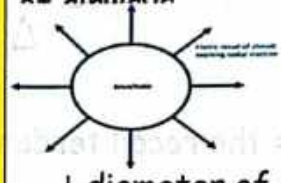




Causes:

A) **Genetic:** deficiency of antitrypsin enzyme.

B) **Acquired:** Chronic irritation & smoking: ↑ release of trypsin enzyme from alveolar macrophages.

Large air ways	Small air ways
Kept open by their cartilages	Have no cartilages Kept open by:
1. Transmural pressure gradient across air ways = air way pressure - IPᵖ o The major factor o ↑ during inspiration → ↑ diameter of airways & ↓ resistance	2. Elastic lateral traction = outward pulling force on sides of airways  ↓ lateral ↓ diameter of airways & ↑ resistance

Compliance & Elastance

Elastic behavior of lungs

Compliance = distensibility & expandibility of lung

= the ratio between: change in lung volume per unit change in the distending pressure (transpulmonary pressure gradient)

$$= \frac{\Delta V}{\Delta P}$$

Elastance = the recoil tendency of lung to its unstretched state.

Compliance is the OPPOSITE of elastance

- ↑ Elasticity → ↓ compliance
- ↓ Elasticity → ↑ compliance

Elastic properties of the lung

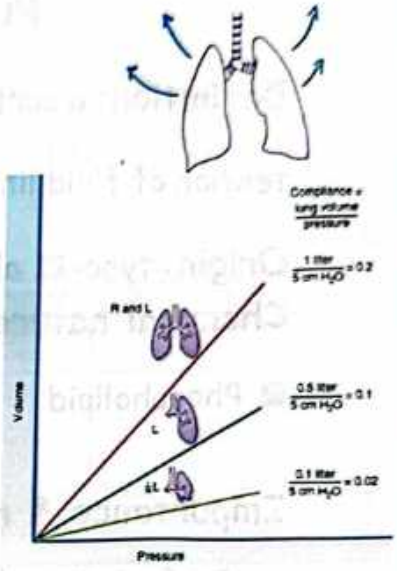
It is caused by 2 factors

Tissue elastic fibers in the lungs	Surface tension.
$\frac{1}{3}$ of elastic properties ◷ Elastin & collagen fibers ◷ Present throughout lung tissue; in alveolar walls, terminal bronchioles & pulmonary capillaries. ◷ Also, present in thoracic wall. ◷ So, both of lung & thoracic wall can be distended and recoil back to their original position	$\frac{2}{3}$ of elastic properties ◷ The fluid lining the alveoli is in contact with air → air-fluid interface → surface tension → ↑ tendency to recoil → ↓ compliance

Compliance

= the degree of lung distensibility

= the change in lung volume per unit change in transpulmonary pressure gradient = $\frac{\Delta V}{\Delta P}$



◷ ↑ Compliance → ↑ expansion of lungs

🕮 Normal value:

- Compliance of lungs = 0.2 L / 1 CmH_2O
- Compliance of lungs & thoracic wall = 0.1 L / 1 CmH_2O

🕮 Factors affecting compliance

Factors ↑↑↑ Compliance	Factors ↓↓↓ Compliance
⊙ Old age (↓ elastic recoil) ⊙ Emphysema (↓ elastic fibers due to breakdown of alveoli) → easy inflation & less pressure is needed for inflation. But: → ↓elastic fibers → : • Collapse of alveoli • ↓ lateral traction ↓ • Difficulty in expiration • ↑ contraction of abdominal muscles • ↑ work of breathing • Active expiration, even during rest	⊙ Pulmonary congestion & Edema ⊙ Pneumonectomy (removal of one lung) ⊙ Pulmonary fibrosis ⊙ Respiratory distress ↓ • Stiff lungs • Difficult inflation • More pressure is needed for inflation

Pulmonary surfactant

Definition: a surface active agent that ↓ the surface tension of fluid lining the alveoli

Origin: type-II alveolar epithelium (granular pneumocytes)

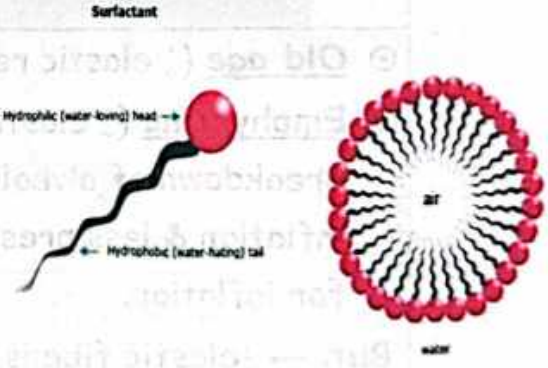
Chemical nature:

- Phospholipid
- Protein
- Calcium

Importance & mechanism of action:

Surfactant molecule has:

- **Hydrophilic part:** facing the alveolar fluid
- **Hydrophobic part:** facing air



So, it forms a lipid surface facing air

→ ↓ air-fluid interface

→ ↓ surface tension of alveoli 2-10 times

→ prevents alveoli from collapse

Functions:

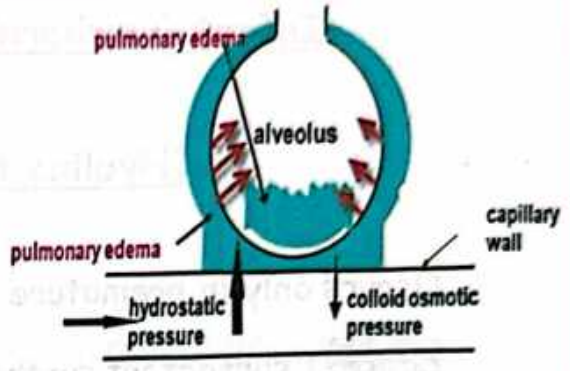
1) ↓ surface tension of the fluid lining the alveoli

2) ↑ pulmonary compliance & lung expansion → ↓ work of breathing

3) ↓ alveolar recoil & prevents their collapse

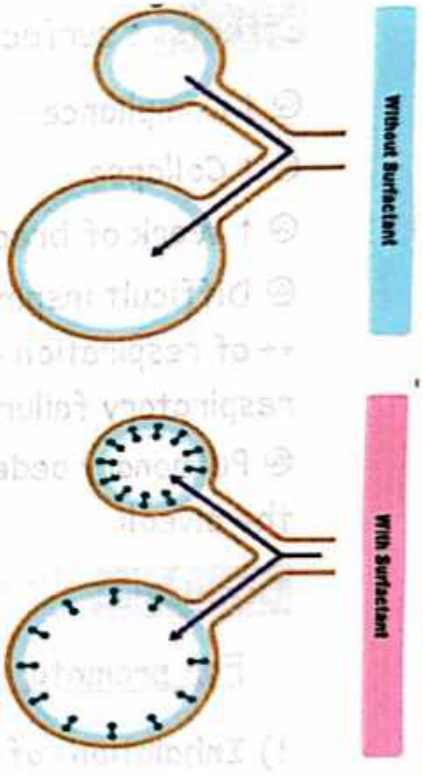
4) Keep the alveoli dry & prevents pulmonary oedema

(accumulation of fluid inside alveoli), BY: preventing collapse of alveoli → prevent development of transudation force from pulmonary capillaries into alveoli



5) Stabilization of alveoli:

- By equalization of the pressure in small & large alveoli
- So, alveoli do not evacuate into each others.



♫ According to the Laplace law:

$$\text{Pressure} = \frac{2T\ (Tension)}{r\ (Radius)}$$

Surfactant: small alveoli > large alveoli

Surface tension: small alveoli < large alveoli

Radius: small alveoli < large alveoli

Pressure: small alveoli = large alveoli

Surfactant deficiency

Infant (newborn) respiratory distress syndrome (IRDS)

[Hyaline membrane disease]

Occurs only in premature babies

Cause: ↓ surfactant synthesis & maturation → ↑ surface tension
Effects: ↑ surface tension →

@ ↓ Compliance
@ ↑ Collapse
@ ↑ Work of breathing
@ Difficult inspiration → ↓ gas exchange (↓ O_2 & ↑ CO_2) → ++ of respiration → ↑ rate of breathing (but inadequate) → respiratory failure & death
@ Pulmonary oedema & accumulation of protenacious fluid in the alveoli

Treatment:

For premature infant

1) Inhalation of synthetic surfactant → ↓ severity of symptoms
2) Positive end expiratory pressure (PEEP) assisted ventilation

For the mother

Glucocorticoids (cortisol) injections before delivery
Aim: ↑ surfactant maturation

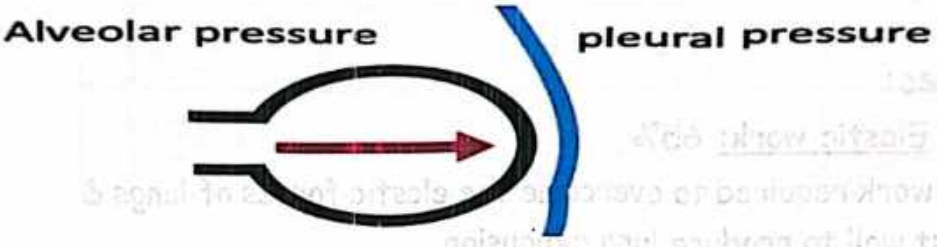
Factors ↑ alveolar collapse:

1) Tissue elastic forces
2) Surface tension

Factors ↑ alveolar opening:

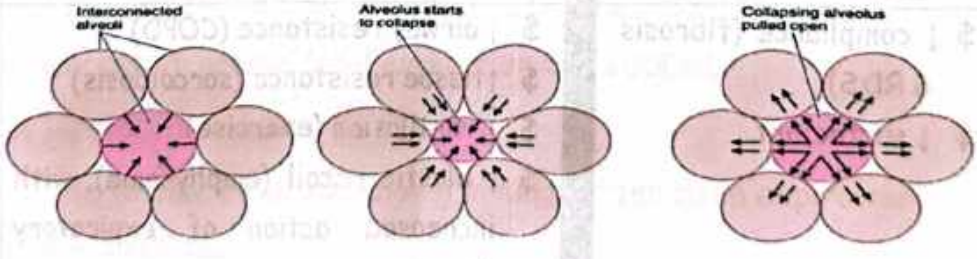
1) Transmural pressure gradient:

Pressure gradient between intralveolar pressure and intrapleural pressure.



2) Surfactant
3) Alveolar interdependence

When an alveolus in a group of interconnected alveoli collapse → stretch of other alveoli → **then** → ↑ elastic recoil of the stretched alveoli → opening of the collapsed alveolus



Work of breathing

Definition: the work performed by respiratory muscles during inspiration

Normally, expiration is passive & no work is done during it.

Normal value: **3% of total body energy expenditure**

- In exercise: it ↑ to 5% of total body energy expenditure
- It becomes ~ 30% in patients with low compliance or with COPD

Types:

1) **Elastic work: 65%**

The work required to overcome the elastic forces of lungs & chest wall to produce lung expansion

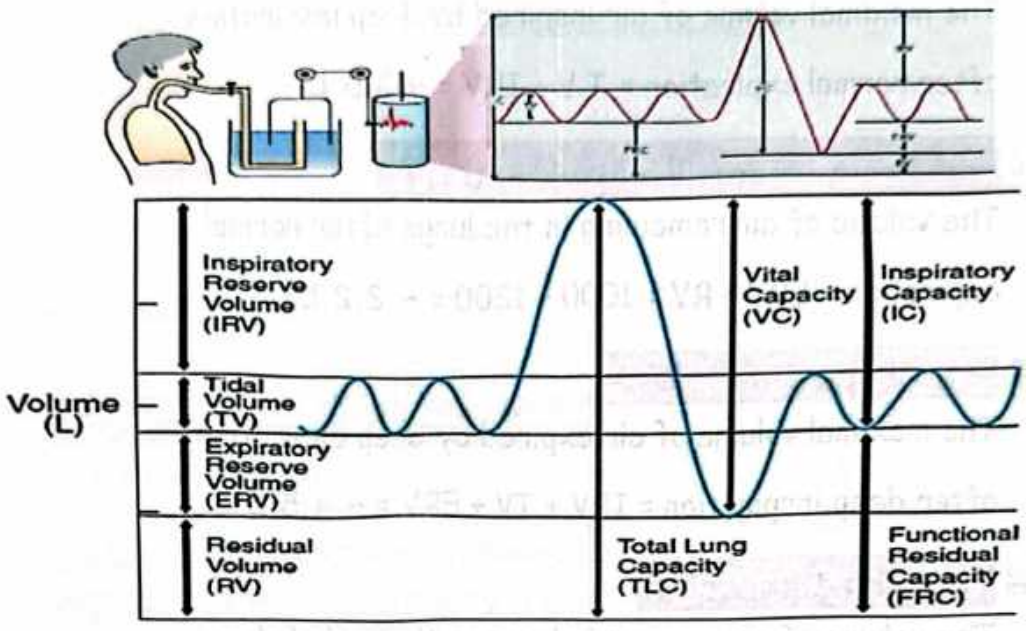
2) **Resistive work (Non- Elastic Work): 35%**

1. **Tissue resistive work (20%):** to overcome the viscosity of lungs & chest wall
2. **Airway resistance (80%):** to overcome airway resistance (minimal)

Work of breathing is ↑ in cases of:

↑ Elastic work	↑ Resistive work
$ ↓ compliance (fibrosis & RDS) $ ↓ surfactant	$ ↑ airway resistance (COPD) $ ↑tissue resistance (sarcoidosis) $ ↑ ventilation (exercise) $ ↓ elastic recoil (emphysema), with increased action of expiratory Muscles

Lung volumes & capacities



⌘ Tidal volume (TV):

The volume of air inspired or expired during normal resting quiet breathing (eupnoea) = ~ 500mL air

⌘ Inspiratory reserve volume (IRV):

The extra volume of air inspired during deep inspiration in excess of normal tidal inspiration = ~3000mL

⌘ Expiratory reserve volume (ERV):-

The extra volume of air expired by deep expiration in excess of normal tidal expiration = ~ 1000mL

⌘ Residual volume (RV):

The volume of air remaining in lungs after deep expiration = ~ 1200mL

⌘ Inspiratory capacity (I.C.):

The maximal volume of air inspired by deep inspiration after normal expiration = T.V + IRV = ~ 3.5 L

⌘ Functional residual capacity (FRC):

The volume of air remaining in the lungs after normal expiration = ERV + RV = 1000 + 1200 = ~ 2.2 L

⌘ Vital capacity (V.C):-

The maximal volume of air expired by deep expiration after deep inspiration = IRV + TV + ERV = ~ 4.5 L

⌘ Total lung capacity:

The volume of air present in lungs at the end of deep inspiration = RV + ERV + TV + IRV = ~ 5.7 L

= IC + FRC = RV + VC

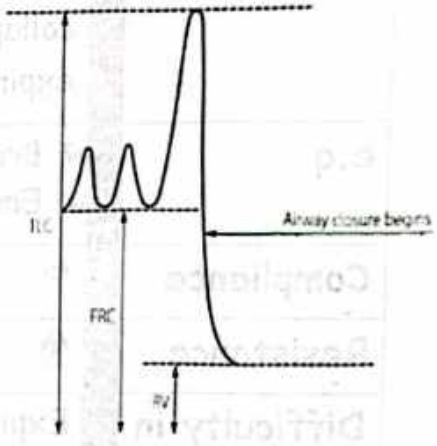
All lung volumes & capacities are:

- Higher in Males than Females by 20-25%
- Higher in large body than small body
- Higher in athletic body than asthenic body
- Measured by spirometer **EXCEPT:**
  - RV
  - FRC
  - TLC

Dynamic lung volumes

A measurement of *lung volumes per unit time*

1) Forced vital capacity
2) Forced expiratory volumes
3) Respiratory minute volume
4) Maximal breathing capacity



1. Forced vital capacity:- (FVC)

= timed vital capacity

Method: deep inspiration, followed by deep expiration & the vital capacity is measured in relation to time

Normal: 4.5 L can be expired in 4-6 secs

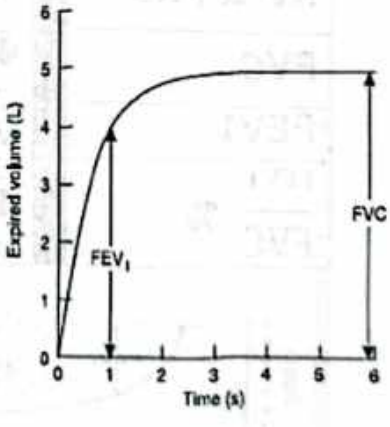
2. Forced expiratory volumes (FEV):

The volume of vital capacity expired in each second

Method: deep inspiration, followed by deep expiration & the vital capacity is measured after each second and expressed as %

$\frac{FEV1}{FVC} X\ 100$ $\frac{FEV2}{FVC} X\ 100$

Normal: FEV1 = 83% FEV2 =94% FEV3 =97%



FEV↓ in:

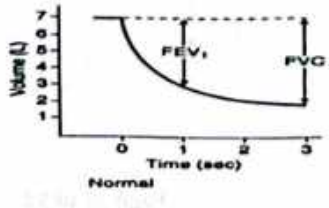
- ☆ Obstructive lung disease
- ☆ Old age

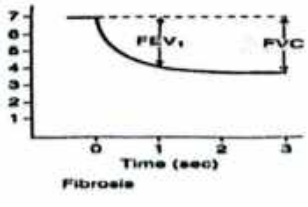
Importance of FVC & FEV:

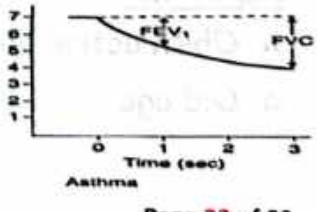
a) Give idea about maximal airflow from lungs.

b) Differentiation between obstructive & restrictive lung diseases.

	Obstructive	Restrictive
Cause	↑ air way resistance → collapse of alveoli during expiration	↓ compliance
e.g	▪ Bronchial asthma ▪ Emphysema	▪ Plum. fibrosis ▪ Rib deformity
Compliance	↑	↓
Resistance	↑	normal or ↓
Difficulty in	Expiration, emptying [air remains in lungs]	Inspiration, filling [↓ air entry in lungs]
TLC	normal or ↑	↓
RV & FRC	↑	normal or ↓
FVC	↓	↓
FEV1	Marked ↓↓↓	↓
$\frac{FEV1}{FVC}$ %	↓ than 80%	normal or ↑







3. Respiratory minute volume (RMV):

= Pulmonary ventilation

Definition: the volume of air inspired or expired per minute

= tidal volume (TV) X Respiratory rate (RR)

Normal = 0.5 L X 12 = 6 L/minute

4. Maximal breathing capacity (MBC)

=Maximal voluntary ventilation (MVV)

Definition: the maximal volume of air inspired or expired per minute using the deepest & fastest breathing

Normal: = 80-170 L/minute in male

= 60-120 L/minute in female

Measurement: by spirometer for 15 sec., & multiply the results X4 (to avoid fatigue & alkalosis)

Importance:

- Never to be normal in lung diseases
- ↑ by: ↑ T.V. & R.R
- It is affected by all factors affecting the vital capacity.

- **Breathing reserve (BR):-** = MVV - RMV
  - It gives idea on reserve of respiratory system.
- **Dyspneic index:** $\frac{BR}{MVV}$ normally ➢ 90%
  - If less than 70% → ↓ reserve → Dyspnea

Vital capacity (VC)

- **Definition:** the maximal volume of air expired by deep expiration, after deep inspiration
- **Measurement:** spirometer
- **Normal value:** 4.5 L
- **Factors affecting:**

1) Condition of muscles of chest (respiratory muscles)

- Athletes have higher VC.
- VC ↓ in: old age , myopathy and in spinal cord lesions

2) Lung compliance: VC ↓ due to decrease compliance in:

- Pulmonary fibrosis
- RDS
- Pulmonary congestion & oedema

3) Range of movement of diaphragm:

Any factor interfere with free mobility of the diaphragm → ↓ VC., e.g:

- Chest deformities & painful conditions
- ↑ abdominal contents: pregnancy, enlarged liver, spleen & ascites
- Recumbent position

- **Importance:** index of physical fitness

Residual volume

- **Definition:** the volume of air remains in lungs after maximal forced expiration
- **Measurement:** by "**Helium dilution principle**" using Helium (inert gas)

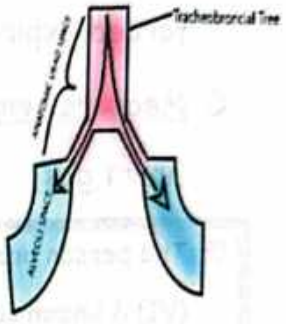
- The person breathes for few minutes from a spirometer of known volume (V1) & known concentration of Helium (C1)
- The helium (known amount = C1 X V1) is dilutes between the: spirometer & lungs (V2) with a new concentration (C2) measured by Helium analyzer
- The known helium amount = C1 X V1 = C2 X V2 =C2 X (V1 + FRC)

- **Normal value:** 1200mL
- **Factors affecting:** ↓ Elasticity → ↑ RV
- **Significance:**

1) Physiological	2) Clinical	3) Medico legal
1) Maintains aeration of alveoli between breathes. 2) Prevents marked changes in O_2 & CO_2 with breathes. 3) Facilitates lung inflation.	♯ Normally, RV = ~ 20-25% of TLC ↑ ♯ RV more than 30% of TLC occurs in COPD	⊙ RV can be expelled by opening of chest wall, but ⊙ Small amount of RV remains (minimal volume) which is sufficient for floatation of lungs after birth.

Dead space (D.S.)

- **Definition:** the volume of air in the respiratory system that doesn't share in gas exchange



- **Types:**

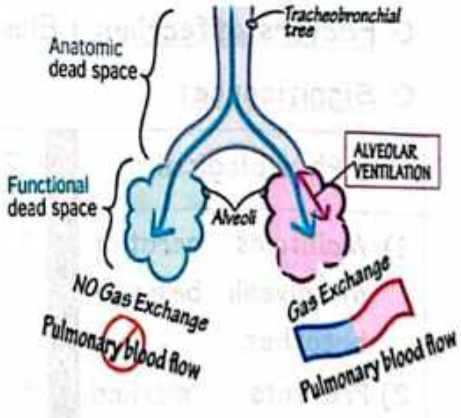
1) **Anatomical D.S. (150mL):**

- Air in conducting airways
- From nose to terminal bronchioles

↑ by	↓ by
# Bronchodilatation	# Bronchoconstriction
# Breathing through tube	# Endotracheal intubation
# ↑ respiratory rate	# Supine posture

2) **Alveolar D.S. (zero):**

Air in non-functioning alveoli (non perfused alveoli = alveoli that don't have a capillary with enough blood)



3) **Physiological D.S.:**

= anatomical + alveolar Normally = anatomical D.S.

So, it is ↑ in conditions of:

- ↑ Anatomical D.S. (by increase respiratory rate)
- ↑ Alveolar D.S. (non-perfused alveoli = alveoli without capillary blood supply e.g. thrombosis or embolism in pulmonary capillaries.

Importance of D.S.:

1) Warms, filters & moistures inspired air

2) Voice production & smell sensation

3) Defensive function by: - Filtration by nose
- Mucous, cilia, coughing & sneezing

4) It causes the difference in composition between alveolar air (has high CO_2 & low O_2) & expired air (lower CO_2 & Higher O_2)

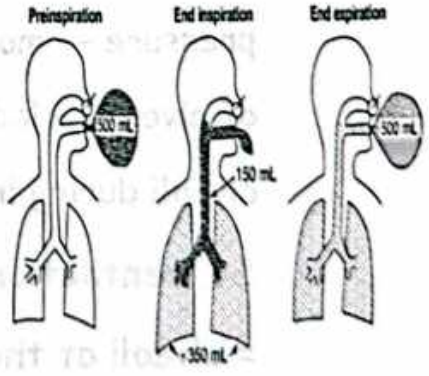
5) It causes the difference between:

★ Pulmonary ventilation = 500 X 12 = 6L/minute &

★ Alveolar ventilation = 350 X 12 = 4.2 L/minute

6) A test of pulmonary function

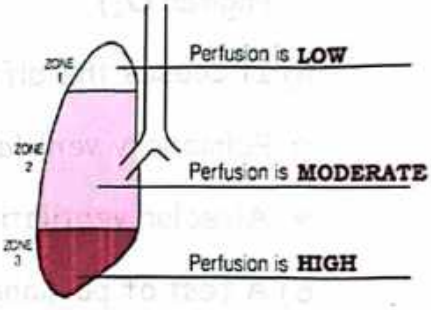
- **In shallow rapid breathing:** most air remains in D.S. → hypoxia
- ↓ pulmonary blood flow → ↑ alveolar D.S → ↑ physiological D.S.
- ↑ breathing rate → ↑ anatomical D.S → ↑ physiological D.S.
- Dead space ventilation = 150 X 12 = 1.8 L/minute
- Pulmonary ventilation = alveolar ventilation +D.S. ventilation
- **D.S. air at the end of inspiration** is fresh air
- **D.S. air at the end of expiration** is alveolar air



Pulmonary ventilation & perfusion

Regional differences in pulmonary perfusion:

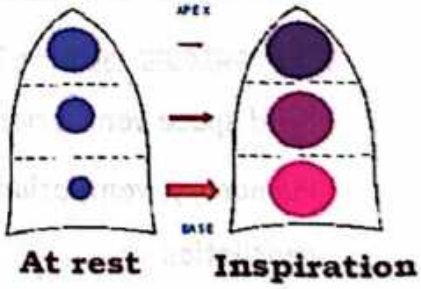
- Due to gravity, blood flow to the base of the lungs is more than apex in upright (erect) posture
- Pulmonary arterial pressure is 3mmHg at apex & 21 mmHg at base → more perfusion to base
- In supine position: blood flow to posterior part of lung is more than anterior parts
- In exercise : pulmonary pressure ↑ → ↑ apical & basal blood flow
- In Hemorrhage → pressure ↓ → underperfusion of the apex



Mean pulmonary arterial pressure: 13mmHg

Regional differences in ventilation:-

At apex: the intrapleural pressure is more ve than the base → ↑ transpulmonary pressure → more expansion → more size of alveoli → less air is moved into & out of alveoli during inspiration.



- So, ventilation at base is higher than apex of the lung.
  = alveoli at the base have: less volume, more compliance

Ventilation/Perfusion $\left[\frac{V_A}{Q}\right]$ ratio

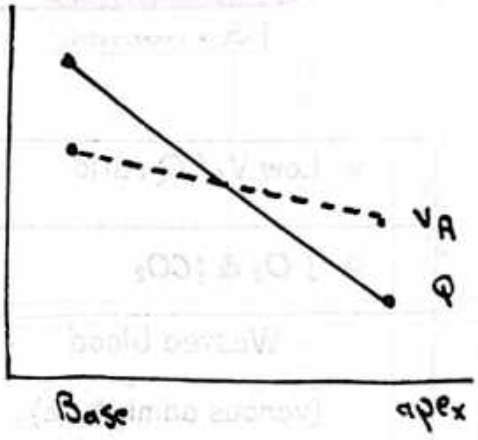
Definition: the ratio between alveolar ventilation (V_A) & pulmonary blood flow (COP = Q).

Normal value: $\frac{4.2}{5} = 0.8$

- For effective gas exchange, V_A/ Q ratio should be ideal in all parts of lungs

BUT, Normally

- V_A/ Q ratio is not the same throughout the lung
- **At apex:** the ratio: 3
- **At base:** ratio: 0.6

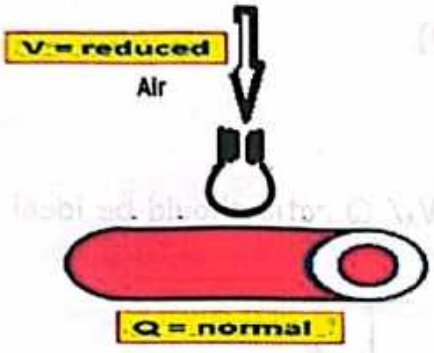
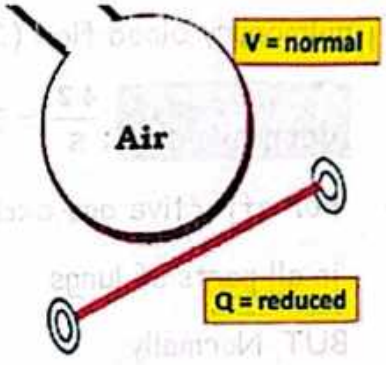
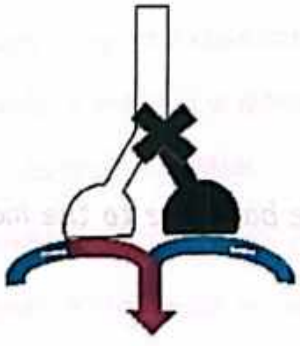


Both of ventilation & perfusion are:

- Not equal all through the lungs
- Higher at base than apex
- Decrease from base to apex

Finally; gas exchange is higher in the base due to the more blood than air.

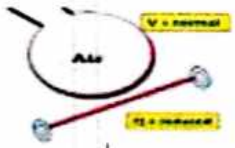
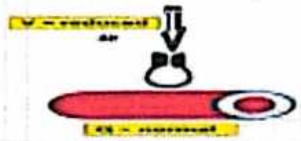
There is a normal physiological mismatching between ventilation & perfusion at:

Lung base	Lung apex
	
❋ Low V_A/ Q ratio	❋ High V_A/ Q ratio
❋ ↓ O_2 & ↑CO_2	❋ ↑ O_2 & ↓CO_2
Wasted blood (venous admixture) (physiological shunt)	Wasted ventilation (physiological dead space)
	

	Physiological dead space	Physiological shunt
Definition	wasted ventilation ☺ air doesn't share in gas exchange.	wasted perfusion ☺ blood leaves the lungs without oxygenation.
Types 1) Anatomical:	🔔 air in conducting airways	🔔 2% of cardiac output 1- Bronchial vessels blood → pulmonary veins 2- Coronary enous blood *(thebesian vessels)* → left ventricle
2) Alveolar:	🔔 High V/ Q ratio = Non-perfused or poorly perfused alveoli ↑ in upright posture (↓ blood flow to upper parts of lungs)	🔔 Low V/ Q ratio = Inadequate ventilation

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CamSc

Physiological compensatory mechanism that minimize the ventilation- perfusion mismatching

Higher V_A/ Q ratio		Low VA/ Q ratio	
More air than blood		More blood than air	
e.g. pulmonary embolism		e.g. airway obstruction	
			
↑ local alveolar PO_2 →	↓ local alveolar PCO_2 →	↓ local alveolar PO_2 →	↑ local alveolar PCO_2 →
relaxation of smooth muscles in pulmonary vessels →	contraction of bronchial smooth muscles →	contraction of smooth muscles in pulmonary vessels →	relaxation of bronchial smooth muscles →
vasodilatation →	bronchoconstriction →	vasoconstriction →	bronchodilatation →
↑ blood flow	local ↓ in ventilation	↓ blood flow	local ↑ in ventilation

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Alveolar PO_2 (100mHg) is LESS than atmospheric PO2 (160mmHg) due to:

1) It becomes saturated with water vapor
2) Only 15% of alveolar air is replaced by atmospheric air during inspiration (350 mL added to 22000 mL)
3) O_2 continuously diffuse into blood

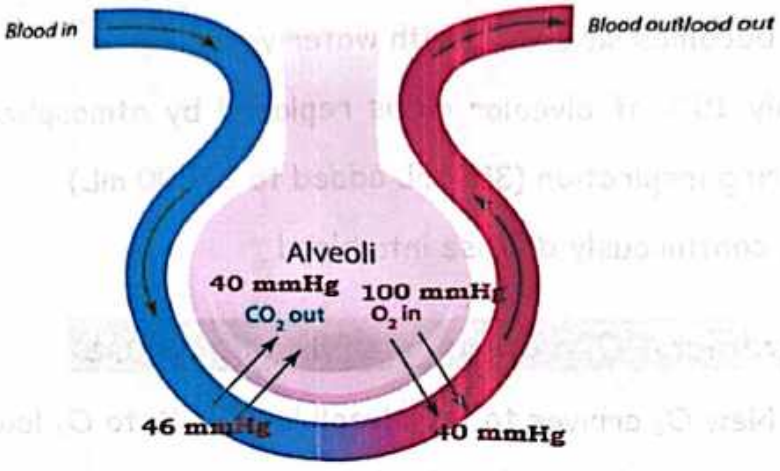
Alveolar PO_2 remains constant because:

- New O_2 arrives to the alveoli is EQUAL to O_2 leaves the alveoli into the blood

Respiratory Gas exchange

- Atmospheric air consists of:
  - @ N_2 =~ 79%
  - @ O_2 =~21%
  - @ CO_2 =~0.04%
- Atmospheric pressure (barometric pressure) = 760mHg at sea level
- Partial pressure of O_2 in air = PO_2 = 760 X 21% = 160mmHg
- Gas exchange between alveolar air (air) & pulmonary capillaries (blood) occurs by simple diffusion
  i.e. from areas of high partial pressure to areas with low partial pressure

Gas exchange at lungs



- Venous blood (poor in O_2 & rich in CO_2) comes in contact with alveolar air (rich in O_2 & poor in CO_2)

a) O_2 diffusion:

- From alveolar air (PO_2 = 100 mmHg) to pulmonary capillary blood (PO_2 = 40mmHg) down pressure gradient
- Blood leaving the pulmonary capillary has PO_2 = 100mmHg

b) CO_2 diffusion:

- From pulmonary capillary blood (PCO_2 46mmHg) to alveolar air
- (PCO_2 40mmHg) down pressure gradient
- Blood leaving the pulmonary capillary has PCO_2 = 40 mmHg

O_2 Reserve:

- Pulmonary venous blood is poor in O_2, but it still has some O_2 that was not used by tissues during resting state.

Diffusion Reserve:

- Pulmonary venous blood passes through the lungs, then back to the heart in 0.75 seconds during resting state.
- O_2 and CO_2 diffusion (equilibrium) occurs in 0.25 seconds only.
- So, there will be always sufficient time for gases diffusion (equilibrium) in the lungs, even during heavy exercise.

CO_2 is always present even in arterial blood:

This helps in:

- Stimulation of respiratory higher centers.
- Regulation of acid-base balance.

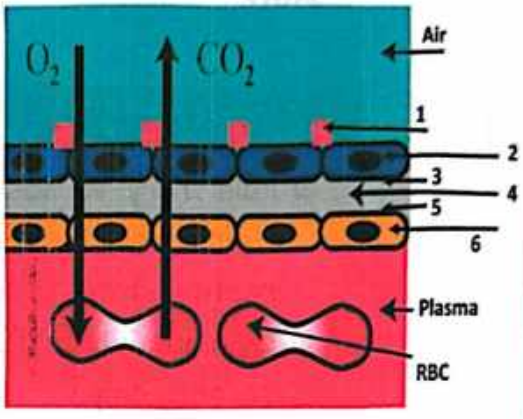
Respiratory Physiology 2026

Alveo-capillary membrane

(= Respiratory membrane)

6 layers:

1) Fluid lining alveoli & surfactant
2) Alveolar epithelium
3) Alveolar basement membrane
4) Interstitial space &fluid
5) Capillary basement membrane
6) Capillary endothelium



Thickness of the membrane: 0.5 u

Surface area of the membrane 50-100 m^2

Factors affecting diffusion of gases:

1) **Pressure gradient:** the gradient (difference) between pressure of the gas in the alveoli (PA) & in the blood capillary (PC)
2) **Surface area** of the pulmonary membrane (A). (70 m^2)
3) **Thickness** of the respiratory system 0.5u (T)
4) **Diffusion coefficient of the gas** (D) = $\frac{\text{Solubility of the gas}}{\sqrt{\text{Molecular weight of the gas}}}$

So,

Rate of gas diffusion = $\frac{D\,A\,(PA-PC)}{T}$

♫ The molecular size of CO_2 is larger than O_2 by 1.4 times.

♫ The solubility of CO_2 is much more than O_2 by 24 times

Net effect:

🕮 The diffusion rate of CO_2 is 20 times faster than O_2

So,

🕮 Respiratory diseases affect O_2 diffusion more than CO_2 diffusion

🕮 Equilibration of O_2 diffusion across the respiratory membrane occurs of at the same time of equilibration of CO_2 diffusion, because O_2 pressure gradient is 60 mmHg while CO_2 pressure gradient is 6 mmHg

Diffusion capacity (DC)

☼ **Definition:** the volume of a gas diffusion from alveoli to pulmonary capillary blood through the respiratory membrane / min / 1 mmHg pressure difference

	O_2 diffusion capacity	CO_2 diffusion capacity
At rest	20 mL / min/ 1 mmHg	400 mL /min/ 1 mmHg
In exercise	60-70mL/ min/ 1 mmHg	1600 mL /min/ 1 mmHg

☼ **Factors affecting DC:**

DC is ↑ in: exercise due to:

↑ Number of opened capillaries & ↑ alveolar expansion

→ ↑ surface area & ↓ thickness

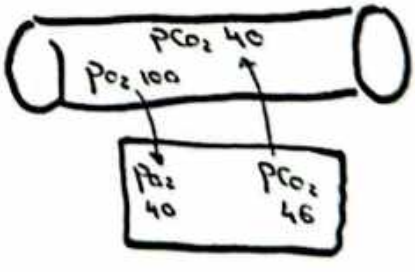
DC is ↓ in:

↑ Thickness of respiratory membrane	↓ surface area of respiratory membrane
♋ Fibrosis ♋ Oedema	♋ Collapse ♋ Emphysema ♋ VA/Q imbalance

Gas exchange at tissues

☆ O_2 diffuses from systemic capillary blood (PO_2 100 mmHg) to tissues (PO_2 40 mmHg).

☆ CO_2 diffuses from tissues (PCO_2 46mmHg) to systemic capillary blood (PCO_2 40 mmHg).



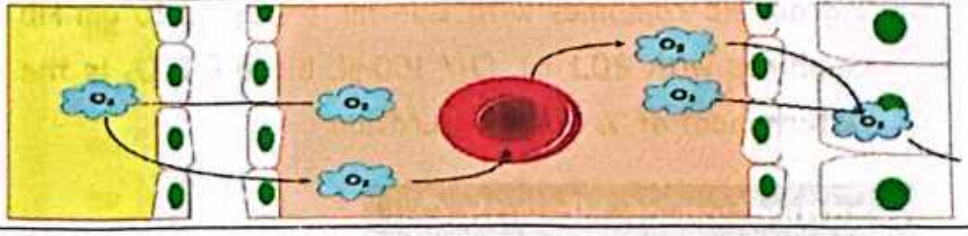
Oxygen transport in blood

- PO_2 in inspired air = 160 mmHg
- In alveoli, inspirec air is mixed with air in the alveoli (FRC) → ↓ PO_2 from 190- 160mmHg to 100-105 mmHg

O_2 in blood:

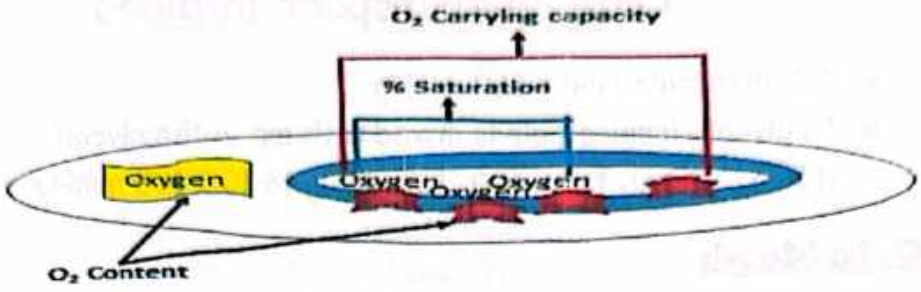
transported in 2 forms

In physical solution	In chemical combination
1.5%	98.5%
Dissolved in water of plasma & RBCs	Binds to the ferrous iron of heme of hemoglobin
0.3 mL / 100 mL arterial blood 0.12 mL/ 100 mL venous blood	19.5mL/ 100 mL arterial blood
Its amount ↑ if PO_2 ↑	One Hb molecule- carries 4 O_2 molecules
It cannot satisfy the demands of tissues	Tissues depend on it



Oxygenation

- Is the rapid (0.01 sec) reversible reaction between each of the 4 iron molecules of Hb with O_2
- The Fe remains in the ferrous state



O_2 content:-

- Total O_2 carried in blood = physical C_2 + chemical O_2
- = 0.3 + 19.5 = 19.8 mL/100mL arterial Blood
- It depends mainly on O_2 bound to Hb.

O_2 carrying capacity of blood:-

- Maximal O_2 carried by Hb when Hb is fully saturated with O_2.
- 1 gram Hb combines with 1.34 mL of O_2. ∎ 15 gm Hb combines with 20.1 mL O_2/ 100mL blood ∎ P O_2 is the determinant of % of Hb saturation.

% saturation of Hb with O_2:-

$$= \frac{\text{O2 bind to Hb}}{\text{O2 capacity}} \times 100$$

- It is a measure of the extent of Hb combination with O_2
- It is determined by PO_2 of blood
- PO_2 (at lungs) → ↑ % saturation of Hb with O_2 = loading of Hb
- PO_2 (at tissues) → ↓ % saturation of Hb with O_2 = unloading

Oxy-Hemoglobin dissociation curve

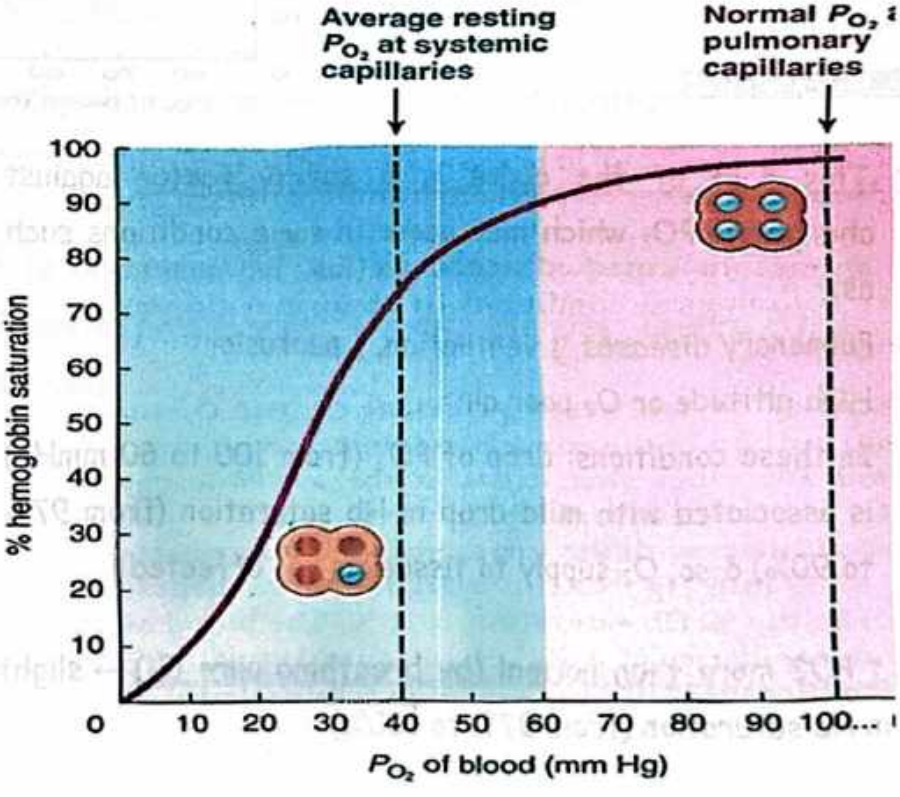
Definition:

- A curve showing the relation between PO_2 & % saturation of Hb (or O_2 content)

Shape:

- Sigmoid (S-Shaped) has 2 parts:

1) Steep part (at low PO_2)
2) Plateau part (at high PO_2)



1. The top Plateau (upper part) of the curve:

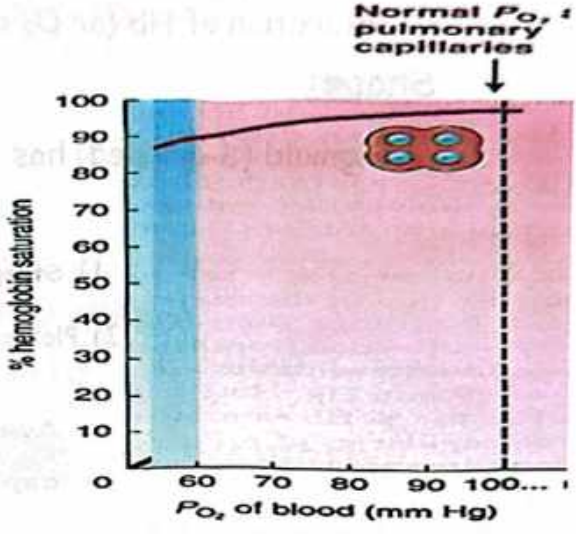
- Represents what happens in the lungs (pulmonary capillaries) where O_2 is high & PO_2 is 60-100 mmHg

→ At PO_2 **100 mmHg**

→ Saturation is **97%**

→ At PO_2 **60mmHg**

→ Saturation is **90%**



Importance:

a) This part of the curve is a safety factor against changes in PO_2 which may occur in some conditions such as:

→ Pulmonary diseases, ↓ ventilation, ↓ perfusion

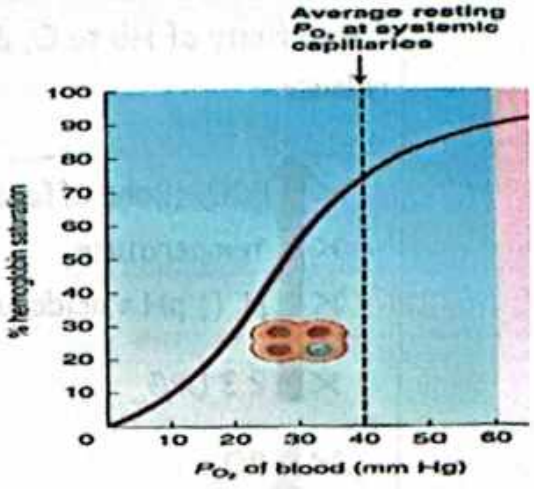
→ High altitude or O_2 poor air

In these conditions: drop of PO_2 (from 100 to 60 mmHg) is associated with mild drop in Hb saturation (from 97% to 90%) & so, O_2 supply to tissues is not affected

b) ↑ PO2 more than normal (by breathing pure O_2) → slight ↑ in Hb saturation (from 97% to 100%)

2. The steep (lower) part of the curve:

- Represent what happens at tissues (systemic capillaries.) where O_2 is low & PO_2 is less than 60 mmHg
  - At PO_2 40 mmHg
  - → Saturation 75%
  - At PO_2 20 mmHg
  - → Saturation is 30%



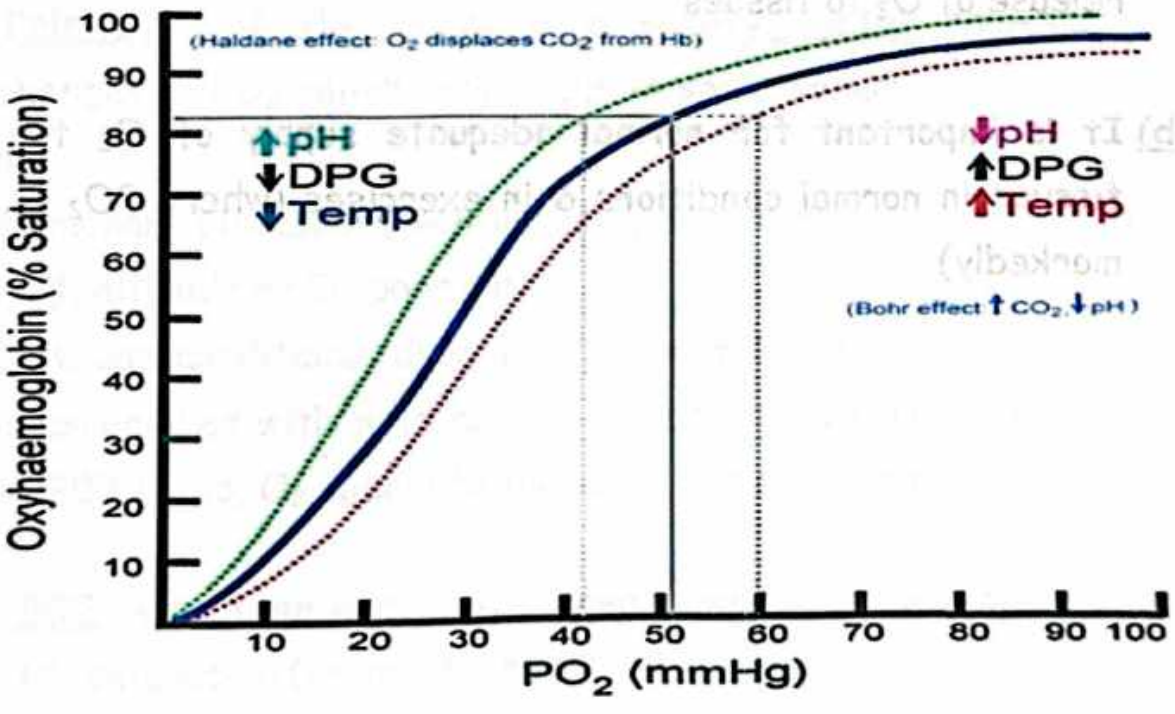
Importance:

a) ↓ PO_2 than 60 mmHg → rapid desaturation of Hb → release of O_2 to tissues

b) It is important for normal adequate supply of O_2 to tissues in normal conditions & in exercises (where PO_2 ↓ markedly)

Factors affecting O_2-Hb curve

Shift to the right	Shift to the left
= ↓ affinity of Hb to O_2 & ↑ O_2 release	= ↑ affinity of Hb to O_2 & ↓ O_2 release
✂ ↑ PCO_2 (Bohr effect) ✂ ↑ temperature ✂ ↑ H^+ (↓ pH = acidosis) ✂ ↑ 2.3 DPG ✂ ↓ PO_2	✂ ↓ PCO_2 ✂ ↓ temperature ✂ ↓ H^+ (↑ pH = alkalosis) ✂ ↓ 2.3 DPG ✂ ↑ PO_2



2, 3 DPG

- Is a substance released from RBCs in anaerobic glycolysis.
- The effect of 2,3 DPG on Hb is very slow (hours).

↑ 2,3 DPG	↓ 2,3 DPG
☼ anemia	☼ stored blood
☼ hypoxia	☼ CO poisoning
☼ exercise	☼ RBC enzyme defect
☼ hormones	

P_{50}

- PO_2 at which 50% of Hb saturated with O_2
- Normal: P_{50} =27 mmHg
- It is an index of Hb affinity to O_2

↑ P_{50} = shift to right

↓ P_{50} = shift to left

- **P_{50} is *inversely* related to Hb affinity to O_2**
- **P_{50} is *directly* related to release of O_2 to tissues**
- **↑ P_{50} occurs in: ↑CO_2, ↑H^+, ↑temperature**

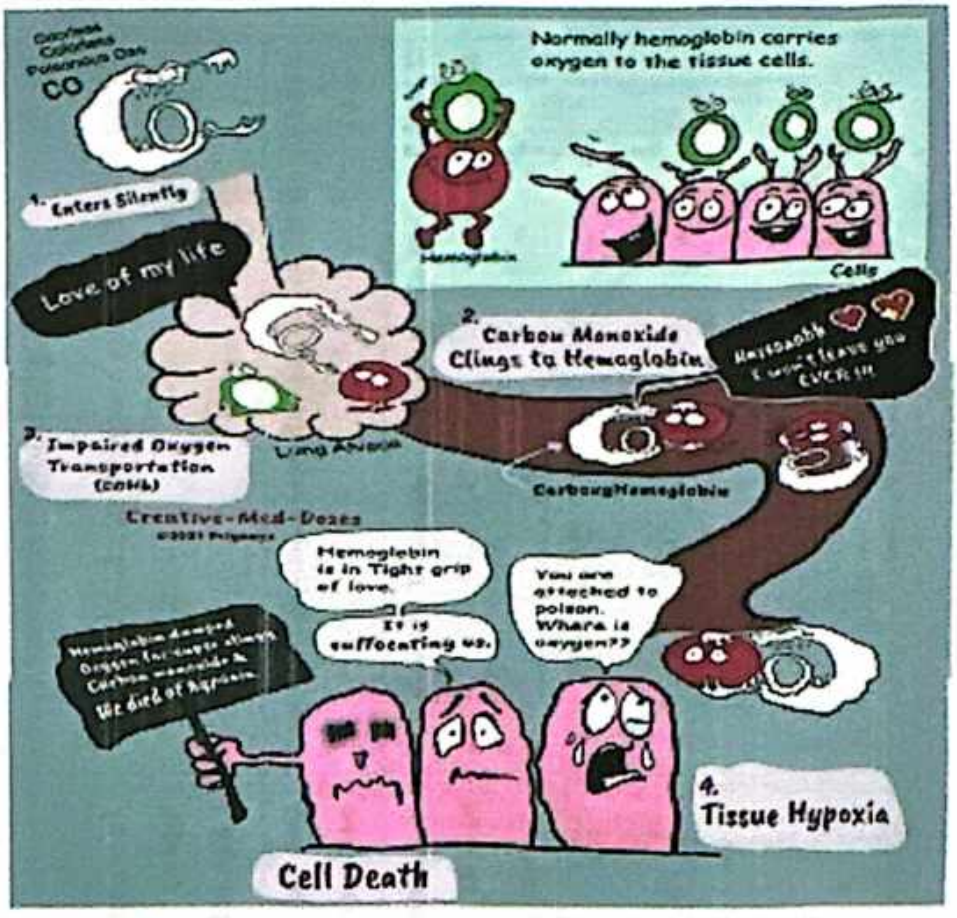
Bohr's effect:

☎ The effect of ↑ PCO_2 & H^+ on O_2-Hb curve

at tissues	at lungs
↑ PCO_2 & H^+ → shift to right	↓ PCO_2 & H^+ → shift to left
↑ release of O_2 from Hb	↑ binding of O_2 to Hb
So, this effect facilitates:	
At tissues: O_2 release from Hb	At lungs: O_2 uptake by Hb

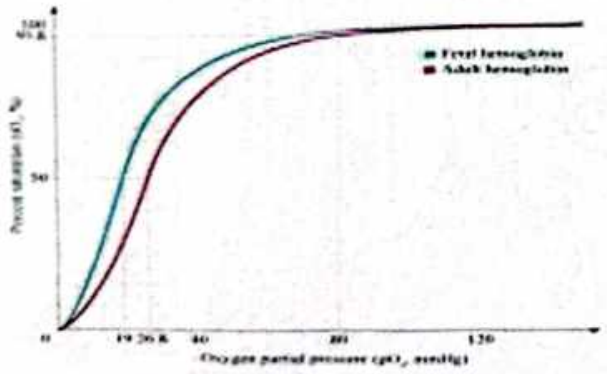
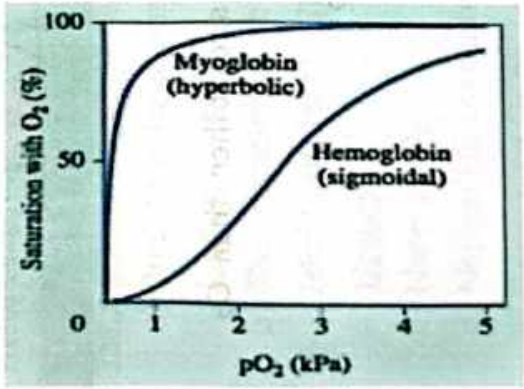
Carbon monoxide (CO) poisoning:

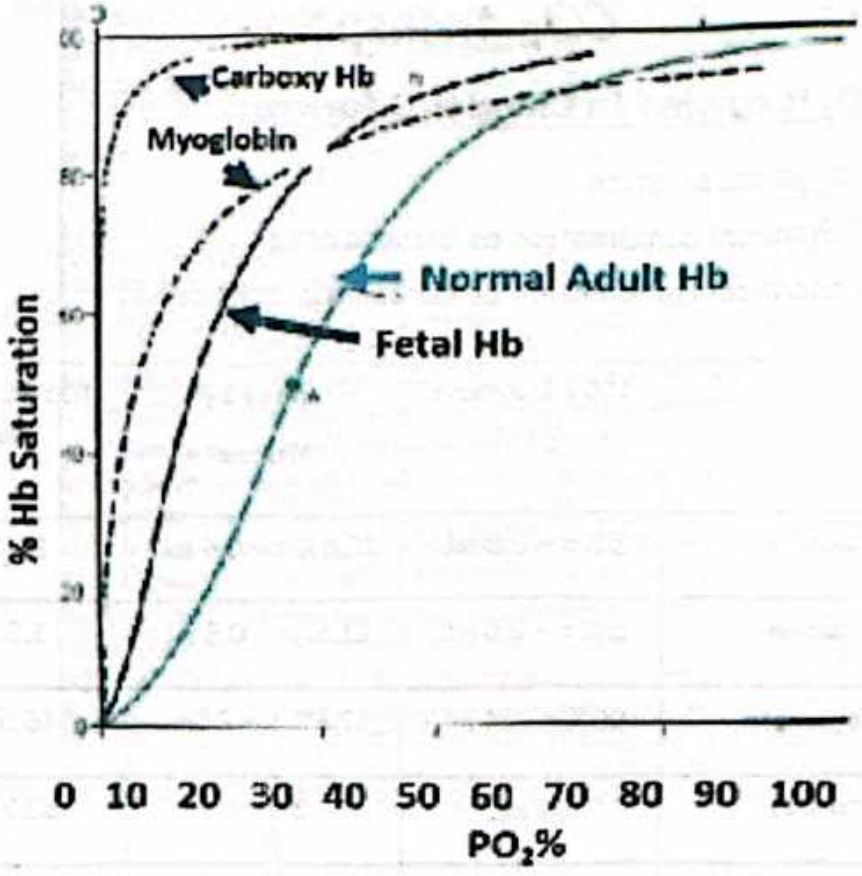
- Binds to Hb at the same O_2 binding site → carboxy Hb (COHb) [or carbon monoxyHb]
- Affinity of Hb to CO is 240 times more than its affinity to O_2
- Very small amount of CO →
  a) Bind to a large amount of Hb (*Interfere with O_2 binding to Hb*)
  b) The remaining Hb decreases the release of O_2 (shift to left) (*Interfere with O_2 release to tissues*)
  c) Normal PO_2
  d) Normal Hb concentration
  e) Decreased O_2 content of blood



CO_2 & H^+:

- Combine reversibly with Hb at sites other than O_2 binding site
- Change the molecular shape of Hb
- ↓ affinity of Hb to O_2
- ↑ release of O_2 to tissues

O_2 dissociation curve of	
Fetal Hb	**Myoglobin**
To the left of adult Hb dissociation curve = high affinity to O_2	To the left of Hb dissociation curve (rectangular hyper.) = high affinity to O_2
Fetal Hb contains 2 Alpha and 2 Gamma polypeptide chains (no B)	One myoglobin molecule carries one O_2 only because it has only one ferrous
2,3 DBG can not act on fetal Hb because it normally acts on B chains 	Myoglobin acts as O_2 store, it releases its O_2 to muscles only in serve exercise (marked ↓ in PO_2) 


Carboxy Hb
Myoglobin
Normal Adult Hb
Fetal Hb
% Hb Saturation
0 10 20 30 40 50 60 70 80 90 100
PO₂%

CO_2 transport

CO_2 is carried in blood in 3 forms:

1) In physical solution
2) In chemical combination as bicarbonate
3) In chemical combination as carbamino-compounds

	100 mL Arterial blood	Added by tissues	100 mL Venous blood
Physical	5% = ~ 2.5 mL	10.8% = ~ 0.4 mL	2.9 mL
Carbamino	5% = ~ 2.5 mL	21.6%= ~ 0.8 mL	3.3 mL
Bicarbonate	90% = ~44 mL	67.6% = ~ 2.5mL	46.5 mL
Total CO_2	49 mL	3.7 mL	52.7 mL
PCO_2	40 mmHg	Tidal CO_2	46 mmHg

Tidal CO_2: the CO_2 given by tissues to 100 cc arterial blood

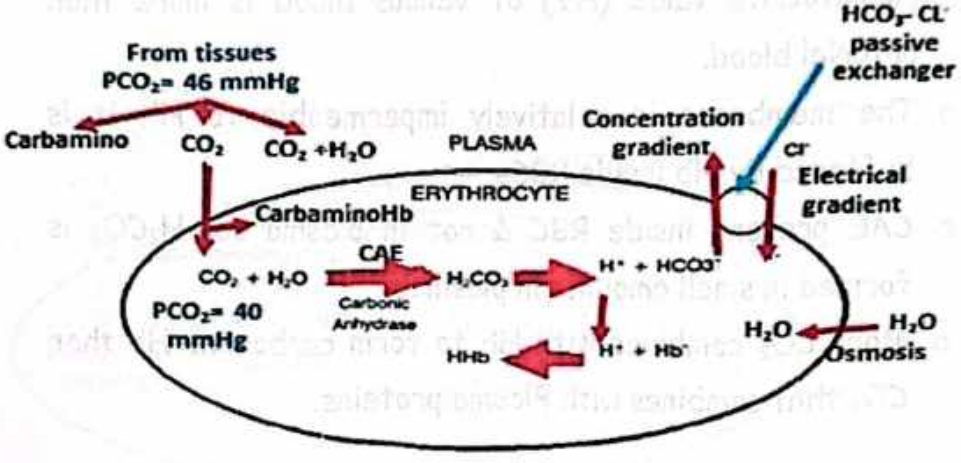
Value: 3.6 - 4mL/ 100 cc (= ~ 3.7 mL)

Venous CO2 = arterial + tidal

During rest, 200mL CO_2 / min transported from tissues to the lungs.

Changes that occur in blood at tissues

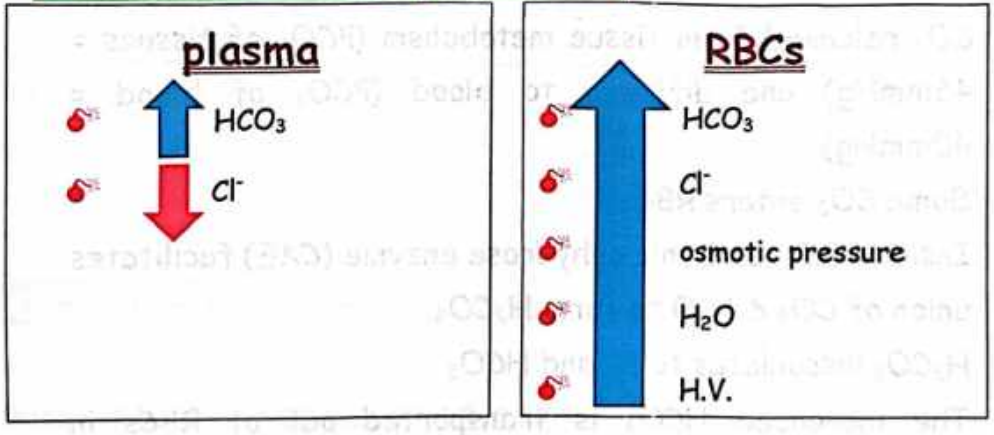
Chloride shift phenomenon



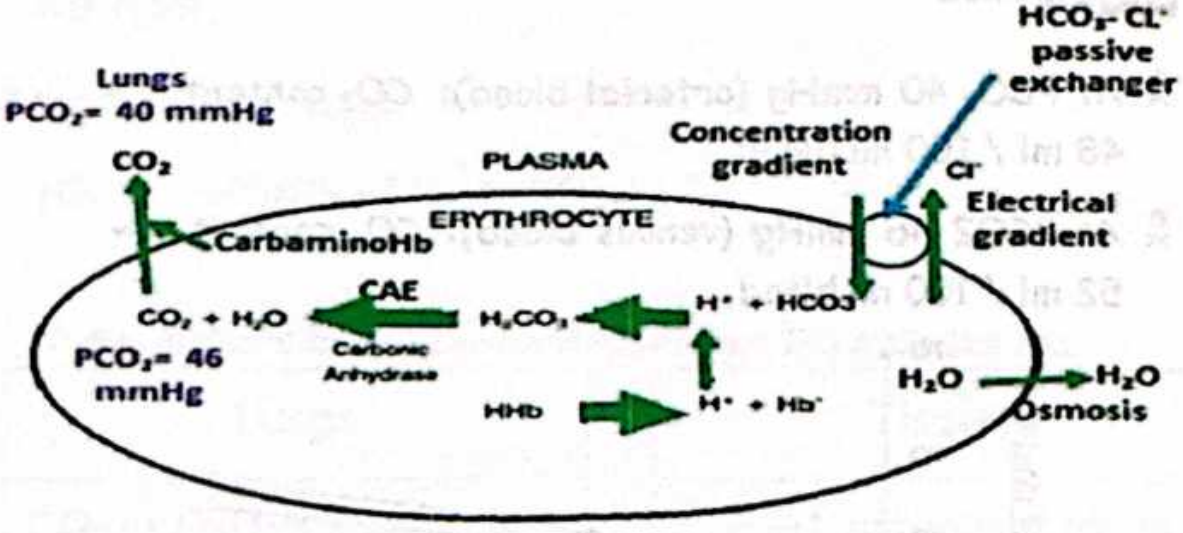
- CO_2 released from tissue metabolism (PCO_2 of tissues = 46mmHg) and diffuses to blood (PCO_2 of blood = 40mmHg)
- Some CO_2 enters RBCs
- Inside RBCs, carbonic anhydrase enzyme (CAE) facilitates union of CO_2 & H_2O to form H_2CO_3
- H_2CO_3 dissociates to H^+ and HCO_3
- The increased HCO_3 is transported out of RBCs in exchange with plasma CL^- by a HCO_3 - CL^- passive exchanger cattier:
  - HCO_3 Moves out by concentration gradient
  - CL^- Moves in by electrical gradient

- The increase CL^- inside RBCS → ↑ RBCs osmotic forces → suction of H_2O to RBCs → ↑ H_2O inside RBCs & ↑ RBCs volume
- Hematocrite value (HV) of venous blood is more than arterial blood.
- The membrane is relatively impermeable to H^+, it is buffered by Hb inside RBCs.
- CAE present inside RBC & not in plasma so, H_2CO_3 is formed in small amounts in plasma.
- More CO_2 combines with Hb to form carbamino Hb than CO_2 that combines with Plasma proteins.

Effects of Cl^- shift phenomenon



- At lung, these processes occur in the opposite direction (CO_2 inside RBCs → wash out to plasma → ↑ HCO_3 in plasma → movement of HCO_3 from plasma to RBCs & movement of CL^- from RBCs to plasma).



- Most of tidal CO_2 is transported as HCO_3 because it is more soluble.
- Tidal CO_2 doesn't cause marked pH changes in venous blood (pH: 7.36) than in arterial blood (pH:7.4) because it is buffered as HCO_3 & carbamino compounds.

Carbonic anhydrase inhibition →

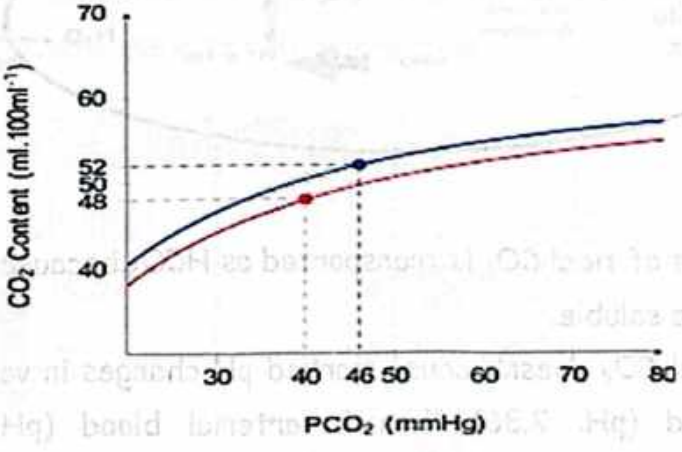
- ↓ Formation of HCO_3 inside RBCs
- ↑ CO_2 in plasma
- ↓ Transport of CO_2 from tissues
- ↑ PCO_2 in tissues

CO_2 dissociation curve

Definition: the relation between PCO_2 & CO_2 content of blood

Shape: linear

- At PCO_2 40 mmHg (arterial blood): CO_2 content is ~ 48 ml / 100 ml blood.
- At PCO2 46 mmHg (venous blood): CO_2 content is ~ 52 ml / 100 ml blood.



Conclusions:

§ Any change in PCO_2 → similar change in CO_2 content

§ Reduced Hb is less acidic than oxyHb → can carry more CO_2 than oxyHb

Effects of O_2 & CO_2 on Hb

a) Bohr's effect:

↑ CO_2 & H^+ → ↑ release of O_2 from Hb to tissues [shift to the right]

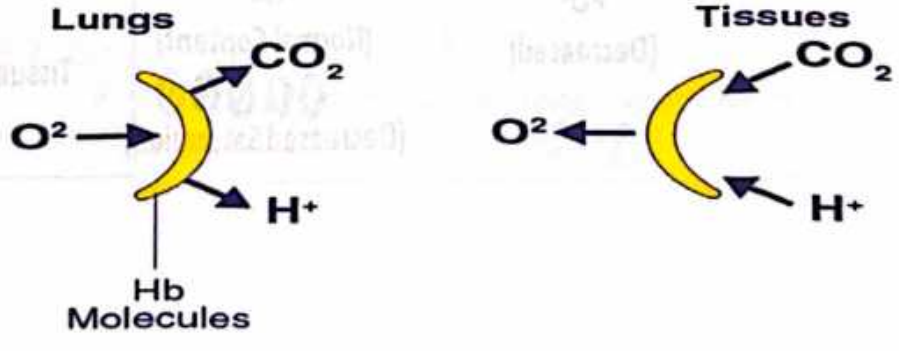
b) Haldane effect:

↑O_2 → ↓ affinity of Hb to CO_2 & H^+

① So, Bohr's & Haldane effect affect Hb affinity at:

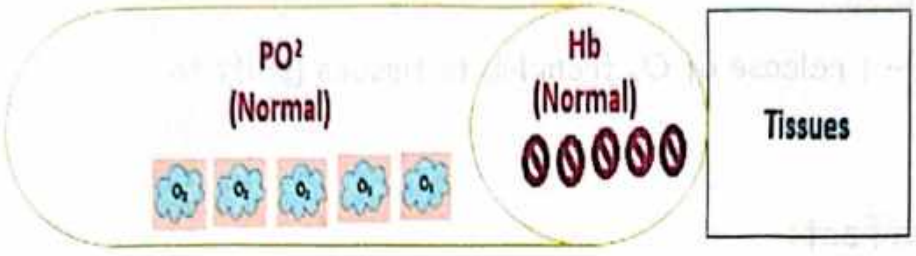
Lungs	Tissues
↑ O_2 → ↓ affinity of Hb to CO_2 & H^+	↓ O_2 →↑ affinity of Hb to CO_2 & H^+
↓ CO_2 & H^+ → ↑ affinity of Hb to O_2	↑ CO2 & H^+ → ↓ affinity of Hb to O_2

The Bohr-Haldane Effect



Hypoxia





Definition: O_2 deficiency at the tissue level

Types & causes:

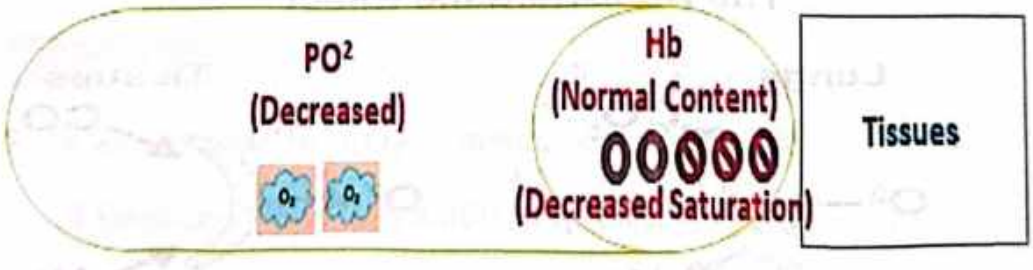
a) Hypoxic hypoxia

↓ Arterial blood PO_2

↓ Hb saturation

↓O_2 content

Definition: Decrease O_2 supply to tissues due to decrease O_2 in arterial blood.



Causes:

1) Extrinsic:

a. ↓ PO_2 in inspired air

b. O_2 poor air in high altitude

c. Hypoventilation due to:

- Respiratory center depression (e.g. by morphine)
- Respiratory muscle paralysis (e.g.poliomyelitis)

2) Pulmonary diseases:

a. Hypoventilation by:

- ↑ resistance
- ↓ compliance

b. Ineffective diffusion:

- ↓ surface area of respiratory membrane (e.g. pneumonectomy)
- ↑ thickness of respiratory membrane (e.g. fibrosis, edema, pneumonia)

c. Abnormal V/Q ratio:

- e.g. emphysema

3) Venous to arterial shunt

- Right to Left shunt (e.g. congenital cyanotic heart disease)

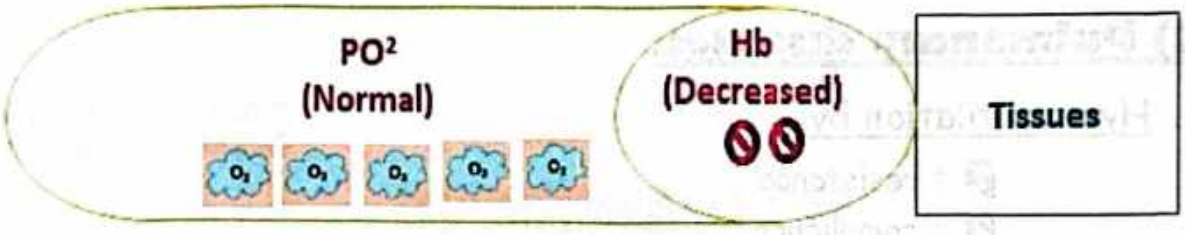
b) Anemia hypoxia

Normal PO_2

↓ Blood O_2 carrying capacity

↓ O_2 content

☎ **Definition:** Decrease O_2 supply to tissues due to decrease Hb amount or function in arterial blood.



☎ **Causes:**

1) Anemia (Qantitative):

☆ Decreased Hb content in blood

2) Abnormal Hb (Qanlitative):

☆ Methemoglobin

3) CO poisoning (Qanlitative)

c) Stagnant (ischemic, Circulatory)

Normal PO_2

Normal O_2 content

↓ Tissue blood flow

Definition: Decrease O_2 supply to tissues due to decrease blood flow to tissues.



Causes:

1) Generalized:

- Shock & Heart failure

2) Localized:

- Vascular spasm or block

d) Histotoxic (cytotoxic)

Normal PO_2

Normal O_2 content

↓ Tissue use of O_2

Definition: Decrease O_2 supply to tissues due to inability of tissues to use O_2.



Causes:

1) Cyanide poisoning
2) Beriberi: Vitamin B deficiency

Effects of hypoxia

- ↓ mental activity drowsiness & coma
- ↓ work capacity of muscles
- Severe hypoxia → Cell death

Clinical Manifestations of hypoxia

1) Headache, easy fatigue, Drowsiness
2) Loss of appetite, nausea, vomiting
3) Tachycardia & Tachypnea

O_2 therapy in Hypoxia

O_2 is beneficial in hypoxic hypoxia in cases of:

- Hypoventilation
- Ineffective diffusion
- High altitude

O_2 is not of value in cases of:

- Hypoxic hypoxia due to venous-Arterial shunt & $\frac{Va}{Q}$ mismatching
- Anemic Hypoxia & abnormal Hb
- Stagnant hypoxia

O_2 under high pressure (hyperbaric) may be used in: histotoxic hypoxia & CO poisoning

Methods of O_2 administration:

- Nasal Cannula
- Mask
- Mechanical ventilator
- Portable





O_2 toxicity

- **Not occur with:** 100% O_2 at 2-3 atmospheric pressure, for less than 5 hours
- **Occur with:** 80-100% O_2 for 8 hours or more

Complications:

1) **On lung:**
   - @ Irritation of airways
   - @ Congestion of nose
   - @ Sore throat
   - @ ↓ surfactant

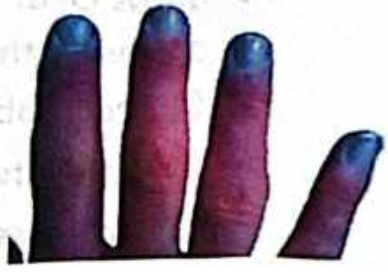
2) **In premature**
   - @ Damage of retina & blindness in neonates (retrolental fibroplasia)
   - @ Bronchopulmonary dysplasia (lung Cysts)

Hyperbaric O_2

- 100% O_2 at high pressure
- Its use ↑ the onset of O_2 toxicity
  - @ Tracheobronchial irritation
  - @ Muscle twitches
  - @ Dizziness, convulsions & coma

- **At 4 atmospheric pressure:** 50% of patients develop symptoms of O_2 toxicity in 30 minutes
- **At 6 atmospheric pressure:** Convulsions occur in few minutes

Cyanosis



⊗ **Definition:** bluish discoloration of skin & mucous membrane due to excess reduced Hb in blood

⊗ **Appears in:** lips, nail beds, air lobes

⊗ **Threshold of cyanosis:** 5 gm reducec Hb /100 mL arterial or capillary blood.

⊗ **Causes:**

1) **Polycythemia:** high Hb content, easy to get more than 5 gm reduced.
2) **Hypoxic hypoxia:** very low PO_2 & O_2 cortent
3) **Stagnant hypoxia:** slow blood flow gives more time for O_2 extraction
4) **Asphyxia:** inability to breath oxygenated air
5) **Moderate cold** → ↓ blood flow → ↑ O_2 extraction → ↑ % of reduced Hb

☺ **No cyanosis in:**

1) **CO poisoning** (cherry red)
2) **Histotoxic hypoxia:** no O_2 consumption by tissues
3) **Anemic hypoxia** (↓ Hb, not easy to get more than 5 gm reduced Hb)
4) **Very cold temperature** → ↓ O_2 consumption by tissues & ↑ Hb affinity to O_2 (shift to left)

Respiratory Physiology 2026

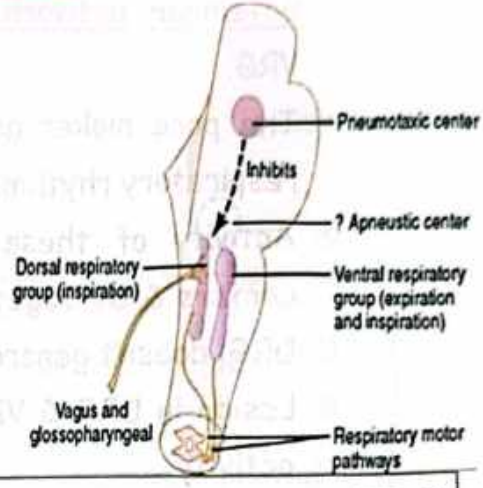
The control of respiration

Respiratory centers

1) Medulla respiratory center:

(the 1ry center)

- In which the automatic rhythmic respiration is generated.
- It consists of 2 bilateral groups of neurons:



Dorsal respiratory group (DRG)	Ventral respiratory group (VRG)
▯ Inspiratory neurons mainly ▯ Acts during normal inspiration	▯ Inspiratory & expiratory neurons ▯ Acts during deep inspiration & forced expiration
✈ DRG → sends impulses to spinal cord → ++ of inspiratory Muscles → inspiration. ✈ DRG → (Stop of impulses) → relaxation of inspiration muscles → expiration.	✈ Deep inspiration: DRG →++ of inspiratory Neurons of VRG → +++ of inspiratory muscles ✈ Deep expiration: DRG →++ of expiratory Neurons of VRG→ ++ of expiratory muscles

- VRG remain inactive during normal breathing.
- VRG is stimulated by DRG (override) during deep ventilation.

Generation of respiratory rhythm:

- The normal respiratory rhythm is generated in Pre-Botzinger network of neurons in the upper end of VRG.
- The pace maker activity of these neurons → normal respiratory rhythm.
- Activity of these neurons is modified by pontine centers & by vagus from lung & airway receptors.
- DRG: doesn't generate the basic breathing rhythm
- Lesion in DRG & VRG : doesn't inhibit the respiratory activity
- Neurons of Pre-Botzinger complex → rhythmic ++ of phrenic nerve → ++ of diaphragm

2) Pontine respiratory center:

a) Pneumotaxic center:-

- In the upper part of pons
- It inhibits apneustic center → makes respiration more rapid & regular (Controls the duration of inspiration). So, it helps in:

1. Stop inspiration at the proper time to increase the rate of breathing.
2. Allowing normal expiration to occur.

b) Apneustic Center:

- In the lower part of pons:-
- It stimulates the medullary center → inspiration
- It is inhibited rhythmically by:

1) Vagus
2) Pneumotaxic center

@ The Pneumotaxic Center dominates over the Apneustic center
@ Vagotomy & damage of pneumotaxic center → slow, deep & prolonged inspiration (apneusis).
@ Damage of Pneumotaxic center alone or Vagi alone→ partial removal of the inhibition on the apneustic center → continuation of breathing but slower & deeper (great tidal volume)

Role of Vagus: [Hering-Breuer reflex]:-

- Over inflation of lungs → ↑ Tidal volume → ++ of stretch receptors in bronchi & lung parenchyma → impulses through vagi nerves → - - of medullary respiratory center → cut off inspiration.
- **Importance:**

1) A protective reflex to prevent overinflation of lungs.
2) No physiological significance in normal conditions.

Regulation of respiration

Chemical	Non chemical
by O_2, CO_2 & H^+ through	chemoreceptors

Chemical control of respiration

- Respiration is stimulated by high PCO_2, low PO_2 & high H^+
- These effects are produced through chemoreceptors
- Chemoreceptors are of 2 types:

Central chemoreceptors

Site: medulla

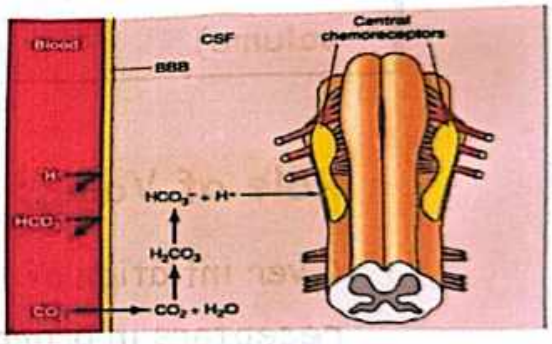
Stimulus: only high PCO_2 but indirectly

High CO_2 in blood → CO_2 crosses the blood brain barrier & enters the medulla

→ $CO_2 + H_2O \rightarrow H_2CO_3$

→ dissociates to HCO_3 & H^+

→ H^+ stimulates the chemoreceptors

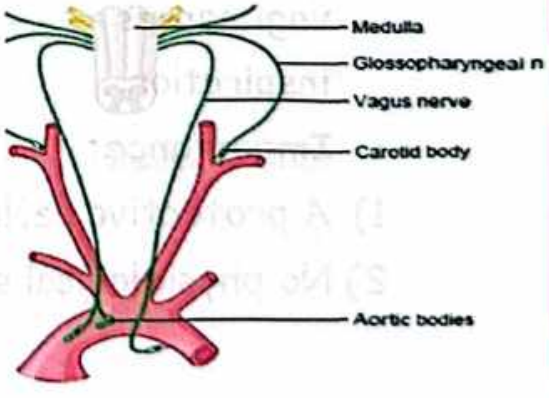


Peripheral chemoreceptors

Site: aortic bodies (in aortic arch) & carotid bodies (in carotid artery)

Stimulus: ↓ in PO_2 less than 60mmHg.

Also ↑ PCO_2 & ↑ H^+

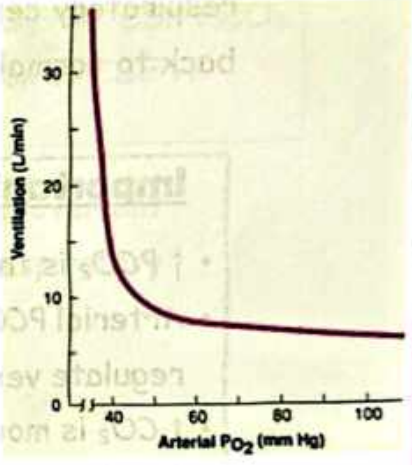


- Peripheral chemoreceptors have very high blood flow.
- So, they depend on dissolved O_2 & so, they are not stimulated if chemical combined O_2 ↓ as in anemia & CO poisoning.
- So, peripheral chemoreceptors respond to PO_2 changes & not to changes in total O_2 content of blood

Ventilatory response to changes in blood gases

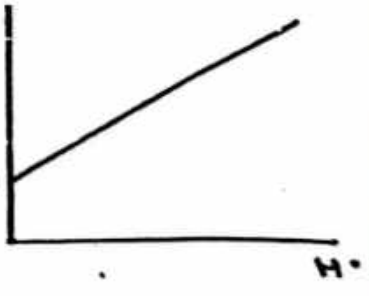
Control of respiration by O_2:-

- ↓ PO_2 (hypoxia) to less than 60 mmHg → Stimulation of peripheral chemoreceptors → ↑ Ventilation.
- **Very low PO_2 →** Depression of respiratory center
- ↑ Ventilation by ↓ PO_2 is a life-saving mechanism used by the body in emergency.



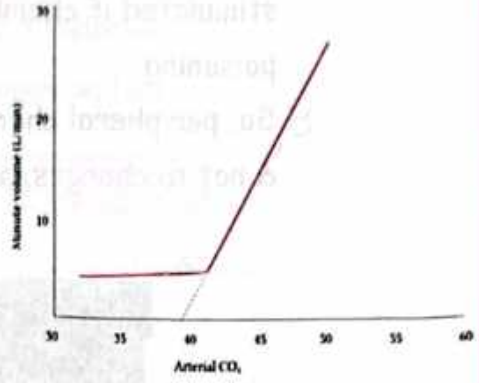
Control of respiration by H^+:-

- ↑ H^+ (acidosis e.g Lactic acidosis) → stimulation of peripheral chemoreceptors mainly → ↑ ventilation.
- ↓ H^+ (alkalosis e.g vomiting) → ↓ ventilation → accumulation of acids.



Control of respiration by CO2:-

- Slight (2-5mmHg) ↑ in PCO_2 (hypercapnia) → Stimulation of central (70%) & Peripheral chemoreceptor → Marked ↑ in ventilation.
- Very high PCO_2 (> 80mmHg) → Depression of respiratory center.
- ↓ PCO_2 (hypocapnia) → ↓ activity of respiratory center → ↓ventilation → accumulation of CO_2 back to normal.



Important Notes:

- ↑ PCO_2 is the 1ry factor that stimulate respiration
- Arterial PCO_2 is the normal most important factor that regulate ventilation (CO2 drive).
- ↑ CO_2 is more powerful in stimulating respiration than ↓O_2.
- Respiratory acidosis (↑ PCO_2) stimulates respiration more than metabolic acidosis (↑H^+).

- **In chronic lung diseases, there are:**

↓ PO_2 below 60mmHg	↑ PCO2
→ ++ of peripheral chemoreceptors → ↑ ventilation	→ adaptation of Central chemoreceptors to the high PCO_2 → loss of sensitivity to CO_2 **This occurs by:** Compensatory mechanism by kidneys to buffer the acidosis →↑ plasma HCO_3 → ↑ CSF HCO_3 → neutralization of CSF H^+

So, in this patient: breathing pure O_2 → improvement of hypoxia → loss of the hypoxic drive to respiratory center → death

Hypoxia is the respiratory drive in:

1) ↓ sensitivity to CO_2 in: chronic lung disease
2) ↓ sensitivity to CO_2 in: very high CO_2 (CO_2 narcosis)
3) Normal CO_2, with ↓O_2 in: high altitude

Respiratory Physiology 2026

Non-chemical control of respiration

= reflexes affecting respiration

= afferents affecting respiration

= neural control of respiration

1. Reflexes from higher centers:

= voluntary control on automatic breathing

e.g voluntary apnea, voluntary hyperventilation, singing & speech

a) From limbic system & hypothalamus:

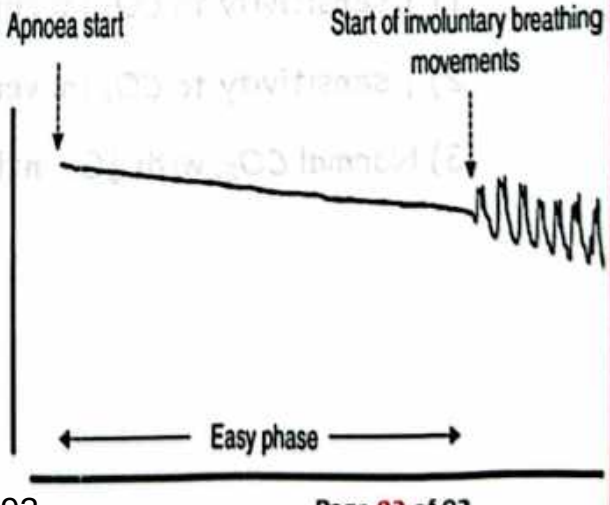
- Pain & strong emotions → ++ sympathetic → ↑ rate & depth of respiration
- Hot → ++ temperature regulating center →++ of respiratory center

b) From cerebral cortex:

Breathe holding test

Voluntary apnea is prolonged by:

1) Breathing pure O_2 before The test.
2) Hyperventilation before The test → washout of CO_2



2. Reflexes from lung receptors:

a) Reflexes from air ways:

- **Stimulus:** irritation of upper & lower airways
- **Receptors:** irritant receptors
- **Response:**

Sneezing: deep inspiration followed by forced expiration with open glottis

Coughing: deep inspiration followed by forced expiration while the glottis is closed→ ↑↑↑ intrapulmonary pressure
Then, the glottis opens suddenly → explosive flow of air out

b) J-receptors (Juxta capillary receptors):-

- **Stimulus:** oedema & congestion (↑ISF) & capsaicin
- **Response:** shallow rapid respiration
- **In severe conditions** → apnea, followed by rapid breathing, hypotension & bradycardia.

c) Pulmonary stretch receptors (Hering- Bruer reflexes)

- **Stimulus:** ↑ volume of inspired air > 1 L
- **Receptors:** high threshold stretch receptors in airways.
- **Afferent:** Vagus
- **Response:** inhibition of respiratory center.
- It is a feedback mechanism that prevent overstretch o lungs.

3. Reflexes from others receptors:

A) **Reflexes from muscles:** (proprioceptors)

a) **Intercostal muscles:** important during normal breathing to determine rate & depth of breathing

b) **Limb muscles:** important during exercise

B) **Reflexes from arterial baroreceptors:**

- ↑ blood pressure → ++ of baroreceptors → - - respiration
- ↑↑↑ blood pressure → Apnea (Adrenaline apnea)

C) **Gut reflexes:**

- Swallowing & vomiting → - - of respiration & closure of glottis

Effect of exercise on respiration

During exercise, there are:

1) ↑ ventilation from 6 L/min to 120 L/min (20 times)

This ↑ in ventilation is:

Abrupt (sudden) then Gradual

- ***In moderate exercise:*** ↑ depth of breathing
- ***In severe exercise:*** ↑ depth & rate of breathing

2) ↑Skeletal Muscles blood flow 30 folds:

- Due to ↑ cardiac output from 5 to 20-35 L/min

3) Arterial Blood Gases:

- ***PCO_2:*** constant or slightly ↓
- ***PO_2:*** constant or slightly ↑

Because the ↑ ventilation is ➢ use of O_2 & the production of CO_2

4) ↑ O_2 extraction by the muscle by:

- ↑ ***Surface area*** of alveoli capillary membrane
- ↑ ***Pressure gradient*** for O_2 diffusion
- ↑ CO_2, H^+, 2.3 DBG & temperature→ ***shift to the right*** of O_2-Hb curve

5) Arterial H^+:

- ***Moderate exercise:*** constant
- ***Heavy exercise:*** → Anerobic metabolism → ↑ lactic acid → ↑H^+ → acidosis (↓ pH)

Mechanisms of ↑ ventilation in exercise

a) Initial ↑ in ventilation at the start of exercise due to:

1) Orders from cerebral cortex (psychic)
2) Impulses from muscles proprioceptors

They are anticipatory feed forward mechanisms

b) Then, gradual ↑ in ventilation occurs due to:

1) ↑ Temperature → stimulation of respiratory center
2) ↑ PCO_2, ↓ PO_2 → fluctuation in their levels → +++ of chemoreceptors
3) ↑ H^+ (in severe exercise) → stimulation of respiratory center
4) ↑ Sensitivity of the respiratory center to CO_2
5) ↑ sympathetic activity → ↑ circulating nor- epinephrine

c) At the end of exercise, ventilation ↓ rapidly & then gradually: May last for 90 minutes.

- During this period O_2 debt occurs

O_2 debt is needed to:

R eplace O_2 of myoglobin
eform ATP & CP
emoval of lactic acid

- Causes of ↑ ventilation during recovery:

The ONLY cause is: ↑ H_+

Effects of high altitude

↓ Barometric pressure

At high altitude:

- ↓ barometric pressure → ↓ alveolar PO2 → Hypoxic Hypoxia
- At 10,000 feet above sea level: → ↓ PO_2 to about 60 mmHg → ++ chemoreceptors → Hyperventilation.
- Above 10,000 feet: marked ↓ in PO_2 → ↓ Hb saturation → marked hypoxic
- The ↓ PO_2 → ++ chemoreceptors
  - Hyperventilation
  - ↑ washout of CO_2 (↓ PCO_2)
  - ↓ arterial H^+ (respiratory alkalosis)

Acute mountain sickness:

Rapid ascend to 10,000 feet or more

- Fatigue & Headache
- Irritability & insomnia
- Nerve dysfunction → dizziness & incoordination
- Nausea
- Vomiting
- Loss of appetite
- Dyspnea
- ↑ HR

Acclimatization to high altitude

= Effects of chronic hypoxia

1) ↑ diffusion capacity by:

- ↑ ventilation→ ↑ surface area of alveo-capillary membrane
- ↑ Pulmonary blood flow→ ↑ surface area of alveo-capillary membrane

2) ↑ O_2 carrying capacity by:

- ↑ Renal secretion of erythropoietin hormone → ↑ RBCs count & Hb content

Disadvantages:

- Marked ↑ in RBCs →↑ Viscosity ↑ Resistance ↑ Heart work

3) ↑ O_2 release to tissues by:

- ↑ 2,3 DBG → shift to the right of Hb curve

4) Cellular compensatory mechanisms:

- ↑ Mitochondria
- ↑ tissue oxidative enzymes
- ↑ myoglobin of muscles

5) ↑ number of opened capillaries in tissues

6) ↑ H^+ excretion →

- Respiratory alkalosis → corrected by excretion of alkaline urine

Pulmonary circulation

☎ **Definition:** a closed circulation, starts from right ventricle of heart → pulmonary artery → pulmonary capillaries (where gas exchange occurs with alveolar air → pulmonary veins → left atrium.

☎ **Aim:** respiratory function (gas exchange).

☎ Characters of pulmonary circulation:

@ Connected to systemic circulation series

@ Low resistance, Low pressure circulation.

@ **Pulmonary blood flow** = cardiac output (COP) of right ventricle = 5 - 6 L/min (equal to COP of left ventricle).

@ **Pulmonary vessels:** short, large diameter, less wall thickness, more distensible, high compliance.

@ Mean arterial pressure: 15 mmHg.

@ Mean capillary pressure: 7 mmHg.

@ Mean venous pressure (in pulmonary veins and left atrium): 2 mmHg in recumbent position.

@ **Reservoir:** accommodate ~ 450 ml blood (9% of blood volume), which return to left side of the heart in emergency (e.g. hemorrhage).

@ **In pulmonary capillaries:** filtration forces ➢ reabsorption forces.

-The excess filtered fluid is absorbed by the very high lymphatic system to keep alveoli dry and prevent oedema.

@ Capillary Transit time: time taken by one RBC to pass the pulmonary circulation = 0.75 sec & 0.3 sec in exercise.

@ During exercise:



@ Cardiac output
@ Pulmonary arterial pressure
@ Capillaries dilatation
@ Opening of more capillaries (recruitment)
@ Pulmonary blood flow

☎ Regulation of pulmonary blood flow:

1) Regulation by hypoxia:

- Local hypoxia: In areas of low Va/Q ratio (less air than blood) → ↓ O_2 (hypoxia) → V.C. of pulmonary blood vessels → decrease blood flow to these alveoli to allow blood to go to the capillaries of the better ventilated alveoli.
- Systemic hypoxia: Also leads to V.C. of pulmonary blood vessels → increase pulmonary arterial pressure.

2) Effect of gravity on renal blood flow:

- Pulmonary circulation is markedly affected by gravity.
- In upright standing position: more blood goes to vessels at the base of the lungs and less goes to vessels at lung apex.

3) Effect of lung volumes on pulmonary vascular resistance:

☺ Pulmonary vascular resistance = resistance of alveolar vessels (vessels around alveoli) + resistance of extraalveolar vessels (other vessels in the whole lungs)

☺ Pulmonary vascular resistance is:

- **Lowest at the end of quiet expiration (FRC)** → maximal pulmonary blood flow.
- **Very high at the end of deep inspiration (TLC):** due to compression of vessels by the distended alveoli → increase resistance of alveolar vessels.
- **Very high at the end of deep expiration (RV):** due to compression of vessels by the intrapleural pressure → increase resistance of extraalveolar vessels.

4) Regulation by autonomic nervous system:

- Pulmonary circulation has rich autonomic innervation.

- Stimulation of cervical sympathetic ganglia → ++ Sympathetic nerves → through α1 adrenergic receptors → V.C. → increase resistance → decrease pulmonary blood flow by about 30%.

5) Regulation by hormones:

Angiotensin II Endothelin
↓
Pulmonary vasoconstriction

Histamine Nitric oxide
↓
Pulmonary vasodilation